



Field Guide to the Geology and Paleontology of Salmon Creek and Ludlowville Falls (Lansing, NY)

A Field Trip Hosted by the Pangea Club at Cornell and Snee Graduate Organization



William C. Hooker¹ & Corey J. Hensen²

¹President, Pangea Club at Cornell (PCC)

²Member, Snee Graduate Organization (SGO)

^{1,2}Department of Earth and Atmospheric Sciences, Cornell University, Ithaca, NY 14853



Table of Contents

Field Trip Itinerary	3
Geology and Paleontology of Salmon Creek and Ludlowville Falls	8
Tectonic setting and paleoenvironment of New York State during the Givetian Stage	8
Lithology and depositional environments present at Salmon Creek and Ludlowville Falls	10
Geologic history of Salmon Creek and Ludlowville Falls after the Givetian Stage	33
The Fossils of Salmon Creek and Ludlowville Falls	34
Summary Chart of the Distribution of Fossils at Salmon Creek and Ludlowville Falls	40
Fossil Identification Guides	43
References	50

Acknowledgements:

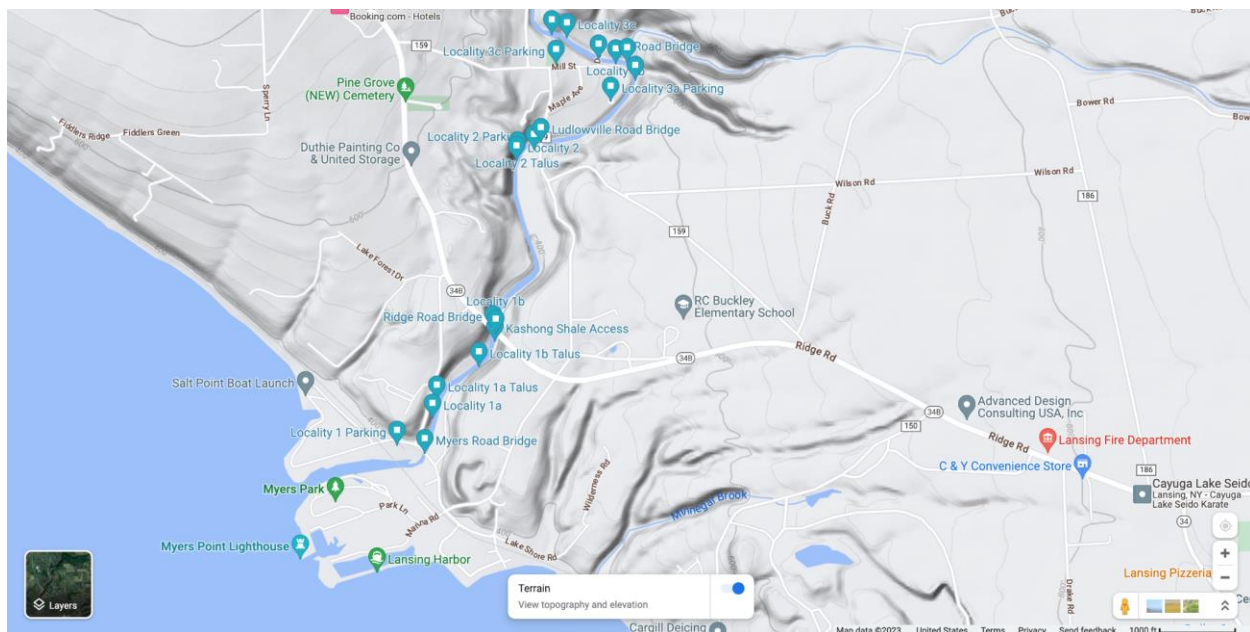
The authors thank Olivia Paschall (President, SGO) for helping to organize and fund this event. PCC thanks its members for their participation as well as those that volunteered to drive and make this field trip possible. Additional thanks to Carlton E. Brett for time spent in the field as well as helpful feedback and discussion.

Anyone using this field trip guidebook is responsible for minimizing environmental impact to Salmon Creek and Ludlowville Falls and agrees not to share, copy, or disperse any of the information contained herein. Please be respectful of wildlife and private property, and leave no trace.





Field Trip Itinerary



Physical map of Salmon Creek.

Please wear appropriate clothing/footwear. Wading through Salmon Creek, sometimes up to waist height, is required to access many of the localities. If collecting fossils, consider bringing plastic bags and/or paper towels, as well as sharpies for labeling locality and stratigraphic information. Note that hammers and chisels are not necessary, as there is ample opportunity to surface collect at most localities. The provided directions to Salmon Creek start at Snee Hall (Department of Earth and Atmospheric Sciences at Cornell University) and end at Ludlowville Falls.

Directions to Locality 1 Parking (Salmon Creek Lower Fishing Access Point) from Snee Hall ($42^{\circ}26'37.7''\text{N}$ $76^{\circ}29'03.8''\text{W}$).

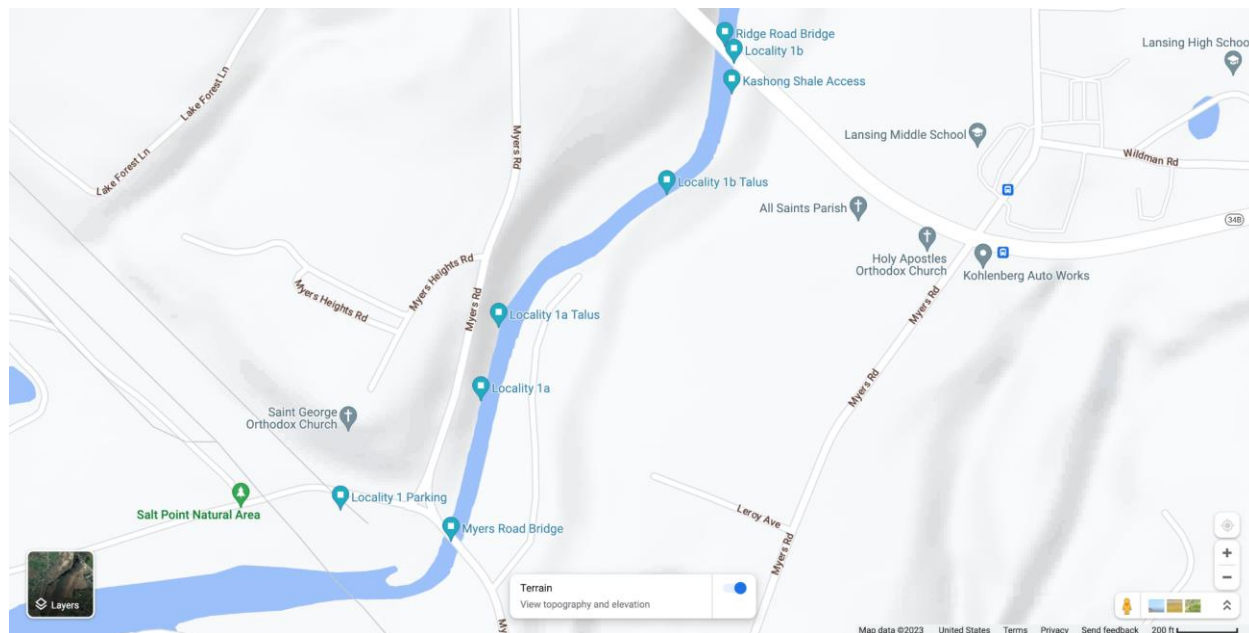
- Take West Ave, University Ave and Lake St to E Shore Dr
- Continue onto E Shore Dr
- Continue on NY-34B N to your destination



Locality 1. Lower Salmon Creek

Spafford Shale, Tichenor Limestone, Kashong Shale, and Windom Shale (*Ambocoelia* Beds)

42°32'23.3"N 76°32'42.1"W



Physical map of Locality 1.

Directions to Locality 1a (Myers Road) from Locality 1 Parking.

- Walk from Locality 1 Parking east along the trail.
- Follow the western bank of Salmon Creek upstream, past the Myers Road Bridge, to the exposed platforms of Spafford Shale, Tichenor Limestone, and Lower Moscow.
- Collect fossils in the Spafford Shale and Lower Moscow (talus) units.

Directions to Locality 1b (Ridge Road) from Locality 1a.

- Follow the western bank of Salmon Creek upstream until directly opposite to Locality 1b Talus. **Be aware of private property and do not trespass.**
- Cross to the eastern bank of Salmon Creek to Locality 1b Talus. **Only cross Salmon Creek in safe conditions at low stage.** Stream stage largely depends on the amount of rainfall that occurs the week prior to the field trip. If safe to cross, be aware of strong currents, deeper pockets of water, and slippery algae on the stream bed.
- Follow the eastern bank of Salmon Creek upstream until immediately below the Ridge Road Bridge.
- Collect fossils in the Kashong Shale and Windom Shale (*Ambocoelia* Beds) units.
- Return to Locality 1 Parking.

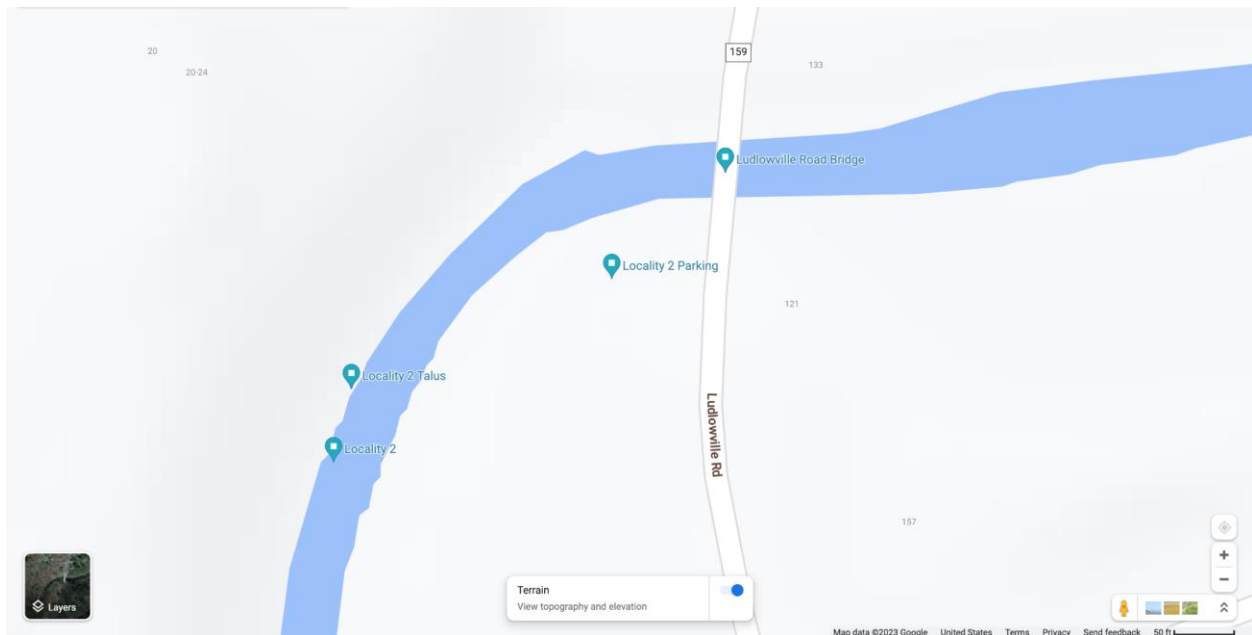


Directions to Locality 2 Parking (Salmon Creek Ludlowville Road Fishing Access Point) from Locality 1 Parking.

- Head east
- Turn left toward Myers Rd
- Merge onto Myers Rd
- Turn left onto NY-34B N
- Turn right onto Ludlowville Rd
- Turn right to stay on Ludlowville Rd
- Destination will be on the right

Locality 2. Middle Salmon Creek

Windom Shale (*Zoophycos* churned beds, Bay View Coral Bed) and Tully Limestone
42°33'00.7"N 76°32'20.8"W (locality) | 42°33'02.2"N 76°32'17.7"W (parking)



Physical map of Locality 2.

Directions to Locality 2 (Ludlowville Road) from Locality 2 Parking.

- Walk from Locality 2 Parking north to Salmon Creek.
- Cross Salmon Creek at its shallowest and least turbid point, typically below the Ludlowville Road Bridge. **Only cross Salmon Creek in safe conditions at low stage.** Stream stage largely depends on the amount of rainfall that occurs the week prior to the trip. If safe to cross, be aware of deeper pockets of water and slippery algae on the stream bed.
- Avoid trespassing once across Salmon Creek.
- Follow the creek downstream along the western bank.



- Collect fossils in the Windom Shale (*Zoophycos* churned, Bay View Coral Bed, and talus) units.
- Return to Locality 2 Parking.

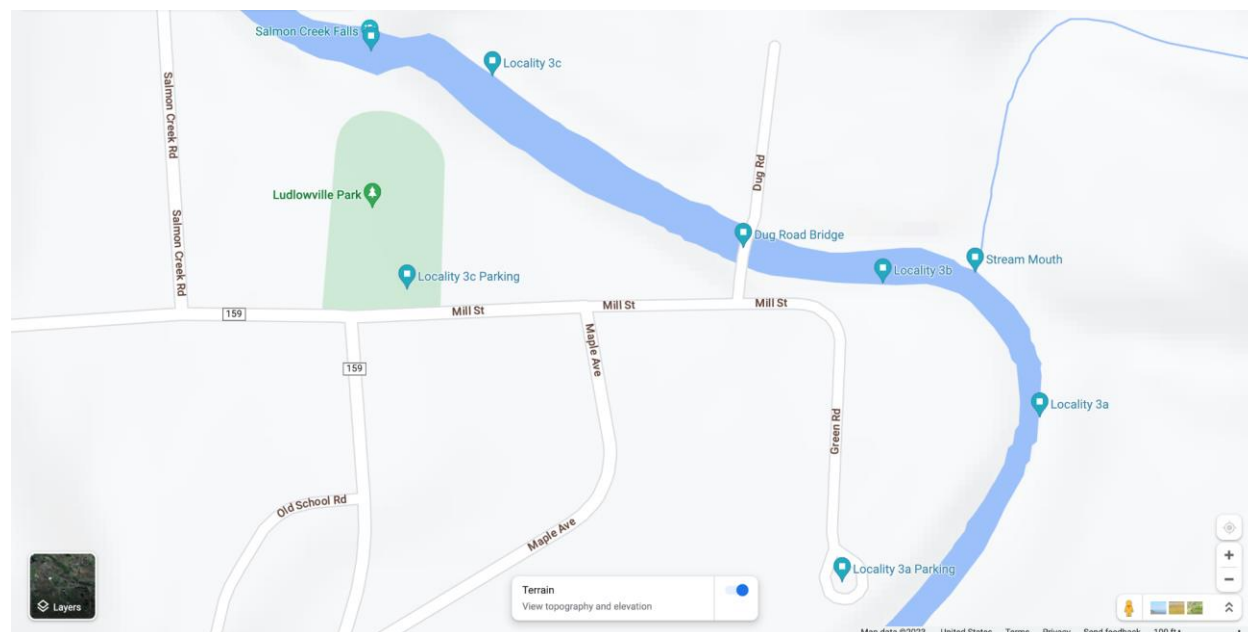
Directions to Locality 3a Parking (Salmon Creek Upper Fishing Access Point) from Locality 2 Parking. Note that additional parking is available at Ludlowville Park; Locality 3 can be accessed in the reverse direction from Locality 3c (Ludlowville Falls) to Locality 3a (Green Road).

- Head west on Mill St toward Ludlowville Rd
- Turn left onto Ludlowville Rd
- Destination will be on the right

Locality 3. Upper Salmon Creek

Windom Shale and Tully Limestone

42°33'13.9"N 76°32'14.2"W



Physical map of Ludlowville Park and Locality 3.

Directions to Locality 3a from Locality 3a Parking.

- Cross to the eastern bank of Salmon Creek immediately from Locality 3a Parking. **Only cross Salmon Creek in safe conditions at low stage.** Stream stage largely depends on the amount of rainfall that occurs the week prior to the field trip. If safe to cross, be aware of strong currents, deeper pockets of water, and slippery algae on the stream bed.
- Follow the eastern bank of Salmon Creek upstream to Locality 3a.
- Collect fossils in the Windom Shale (concretionary horizon) units.

Directions to Locality 3b (Dug Road) from Locality 3a.



- Follow the eastern bank of Salmon Creek north until reaching the stream mouth. **Be aware of private property and do not trespass.**
- Cross to the southern bank of Salmon Creek at the stream mouth. **Only cross Salmon Creek in safe conditions at low stage.** Stream stage largely depends on the amount of rainfall that occurs the week prior to the field trip. If safe to cross, be aware of strong currents, deeper pockets of water, and slippery algae on the stream bed.
- Follow the southern bank of Salmon Creek upstream to Locality 3b.
- Collect fossils in the Windom Shale (Bear Swamp Beds) units.

Directions to Locality 3c from Locality 3b.

- Follow the southern bank of Salmon Creek upstream until immediately across from Locality 3c.
- Cross to the northern bank of Salmon Creek. **Only cross Salmon Creek in safe conditions at low stage.** Stream stage largely depends on the amount of rainfall that occurs the week prior to the field trip. If safe to cross, be aware of strong currents, deeper pockets of water, and slippery algae on the stream bed.
- Collect fossils in the Windom Shale (*Ambocoelia* Beds) units.
- Cross to the southern bank of Salmon Creek.
- Follow the southern bank of Salmon Creek upstream until reaching Ludlowville Falls. Enjoy the view!
- Return to Locality 3a Parking by following the trail from Ludlowville Falls to Ludlowville Park, and then following Mill St east to Green Rd.

Directions from Locality 3c Parking back to Snee Hall.

- Take Brickyard Rd and NY-34B S to NY-34 S/E Shore Dr
- Turn right onto NY-34 S/E Shore Dr
- Continue on Lake St to your destination

End of Field Trip.



Geology and Paleontology of Salmon Creek and Ludlowville Falls

Tectonic setting and paleoenvironment of New York State during the Givetian Stage

During the Givetian Stage (late Middle Devonian), the Acadian Orogeny¹ occurred as a result of collision between the eastern margin of North America with the Avalon terrane. The uplift of the Acadian Mountains created an area of thickened crust to the East, driving flexural subsidence and the creation of a foreland basin (Fig. 1). New York State was situated in this basin during the Givetian, and an epeiric sea² was formed as seawater filled it (Fig. 2). Sediments shed from the Acadian Mountains formed the Catskill Delta, a thick section of marine to non-marine sediments spanning the Middle and Late Devonian of New York. The sedimentary rocks exposed at Salmon Creek (Lansing, NY) represent the offshore equivalents of this deltaic sequence within the upper part of the Hamilton Group. These rocks are known to contain a rich fossil fauna that record life in this ancient sea (Wilson, 2014). They also contain a record of sea-level change, evident in cyclic packages of rock bound by surfaces of erosion (Brett et al., 2012). These features are readily observable in the field in a series of discontinuous outcrops exposed along the gorge walls of Salmon Creek. An overview of several key stratigraphic and paleontological features is provided below. Key stratigraphic terms and a examples of common fossils are provided in the Appendix.

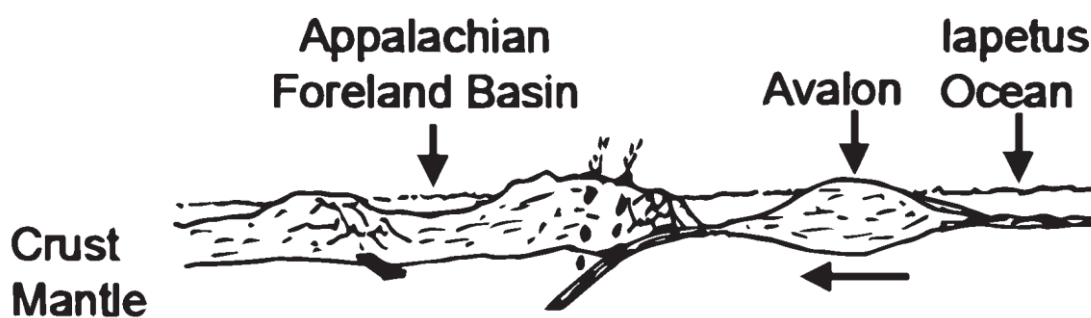


Figure 1. Tectonic setting of New York State during the Middle Devonian Period (Wilson, 2014).

¹ the formation of a mountain range through the compression and folding of Earth's crust

² a sea extending over Earth's continental crust

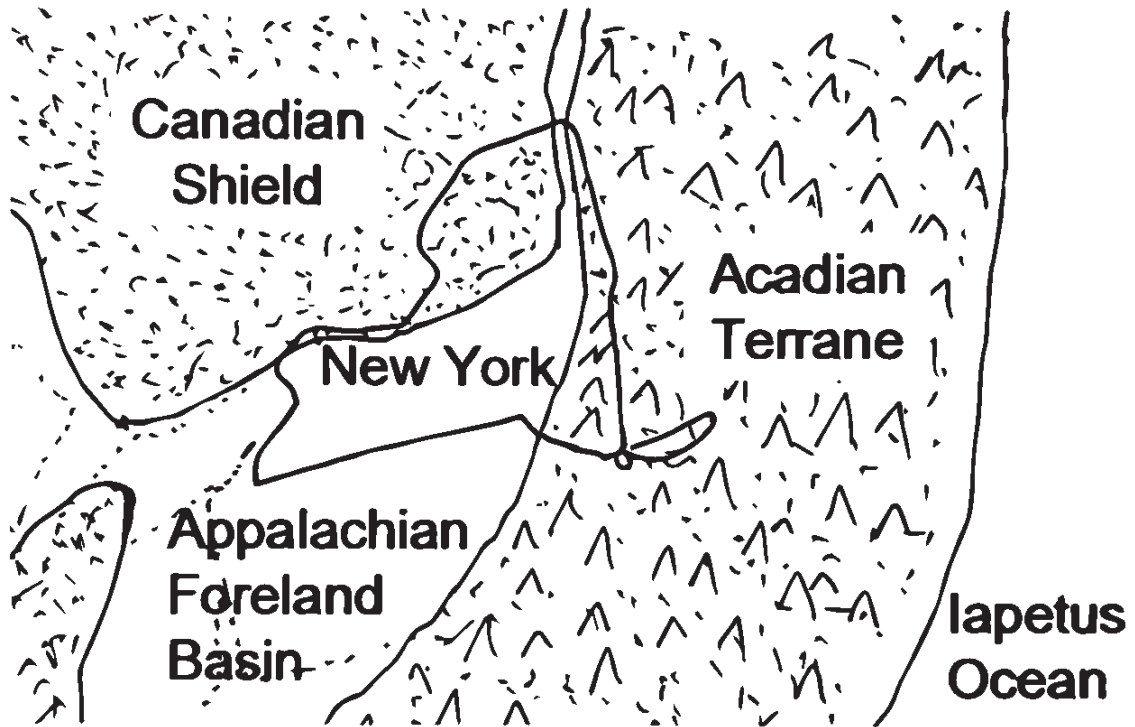


Figure 2. Paleogeography of New York State during the Middle Devonian Period (Wilson, 2014).

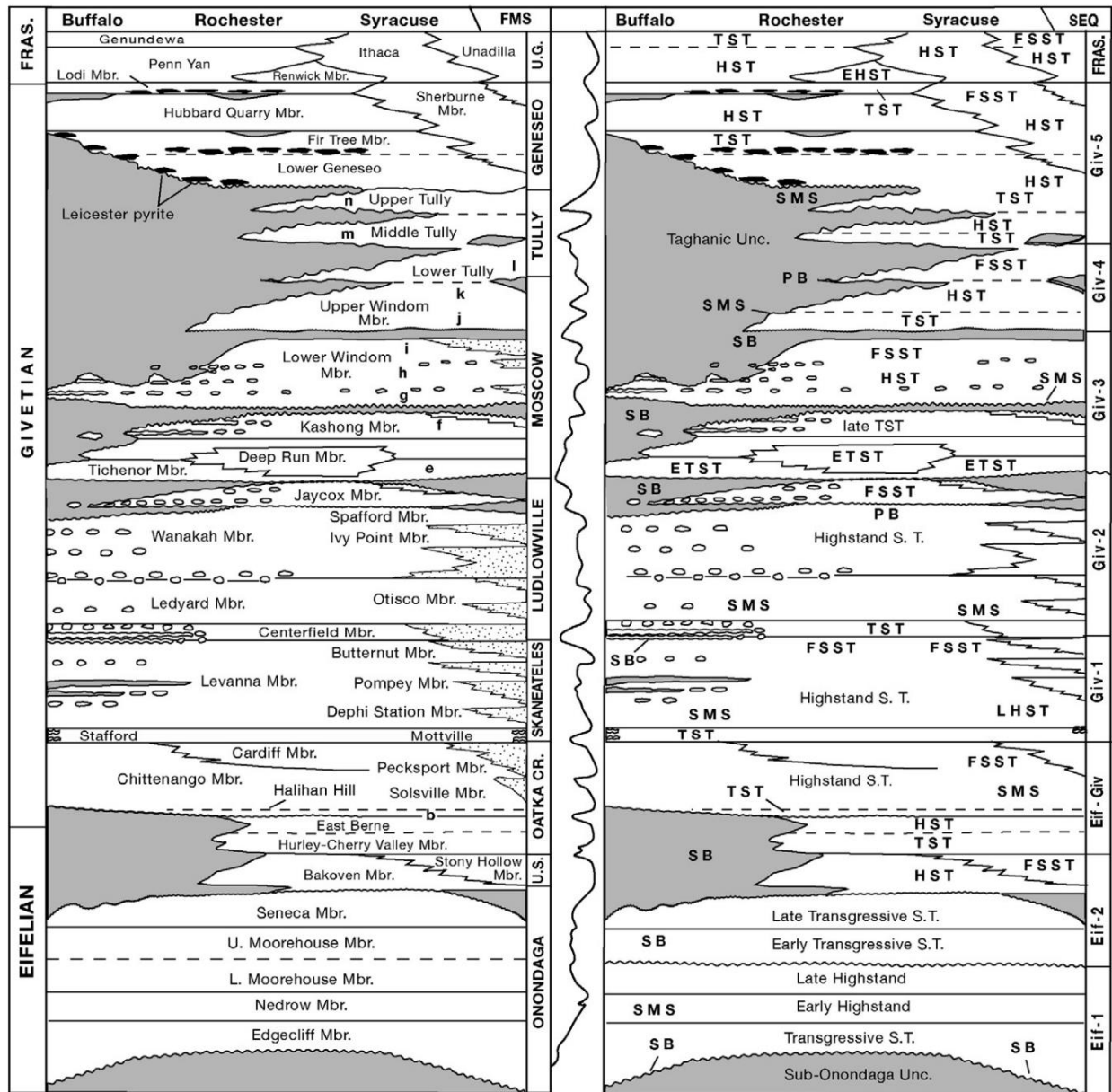


Figure 3. Devonian lithostratigraphy (left) and sequence stratigraphy (right) of New York State (from Brett et al., 2012). Strata exposed at Salmon Creek includes units of the Ludlowville and Moscow Formations of the Hamilton Group. At Locality 1a, Spafford Shale, Tichenor Limestone, and Kashong Shale are present. At Localities 2 and 3, Windom Shale and Tully Limestone are present. Inferred trends in relative sea level are illustrated by the middle curve. See Appendix for sequence stratigraphic terminology.



Stratigraphic and paleontological features at Salmon Creek and Ludlowville Falls

Locality 1. Lower Salmon Creek

Spafford Shale, Tichenor Limestone, Kashong Shale, and Windom Shale

42°32'23.3"N 76°32'42.1"W

Locality 1a. Myers Road

Spafford Shale and Tichenor Limestone

42°32'26.9"N 76°32'35.8"W



The Spafford Shale Member of the Ludlowville Formation is composed of gray, fossiliferous siltstones. This unit contains many brachiopod fossils, typically found in discontinuous lags (stringers) of shell material. Common spiriferid brachiopods include *Ambocoelia umbonata*, *Athyris spiriferoides*, and *Mucrospirifer mucronatus*. Several subclasses of bivalves are also present in the Spafford Shale, including anomalodesmatids, heterodontids, palaeoheterodontids, and pteriomorphs. Large ovate concretions are abundant, but do not often contain well-preserved fossil material. However, split concretions will occasionally reveal calcitic assemblages of spiriferid brachiopods. Note that the Jaycox Shale is not present at Salmon Creek. This unit, only present in sections further to the west, is about two meters thick and generally represents deepening water depth, a decrease in

oxygen content, and a decrease in relative abundance of marine faunas (Brett et al., 2012). Together, these units represent the upper part of sequence Giv-2, a major sea level cycle recognized throughout the Middle Devonian of North America (Figure 3; Brett et al., 2012).

Inset left: exposures of Spafford Shale, Tichenor Limestone, and Lower Moscow at Locality 1a.



Inset right: A typical lag in the Spafford Shale at Locality 1a, containing calcitic valves of the strophomenid brachiopod *Devonochonetes coronatus*. Notice that a second shell stringer located just below the one in focus.

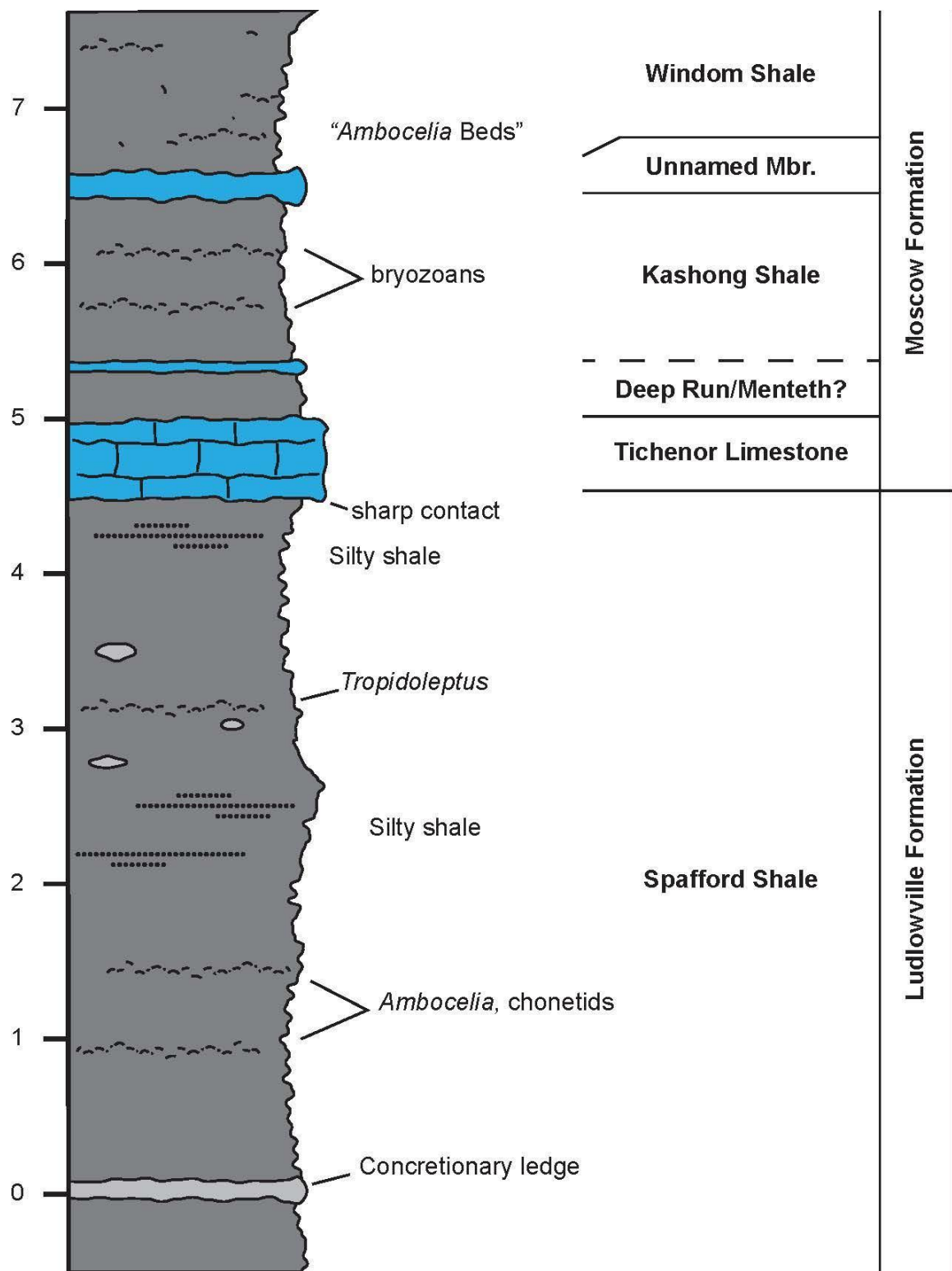


Figure 4. Measured section of the Ludlowville-Moscow contact interval at Locality 1. Scale in meters.



Typical bivalves of the Spafford Shale at Locality 1a. Top left: right valve of the pteriomorph *Pseudaviculopecten princeps*. Top right: both valves of the palaeoheterodontid *Goniophora hamiltonensis*, preserved three-dimensionally. Bottom left: a “butterfly specimen” (i.e., valves spread open) of the palaeoheterodontid *Modiomorpha mytiloides*. Bottom right: left valve of the anomalodesmatid *Grammysioidea alveata*.



A split ovate concretion reveals the calcitic valves of spiriferid brachiopods.



A rare instance of a pyritized partial phragmocone of the orthocone nautiloid *Spyroceras nuntium*. Though this specimen was collected as float, it likely is a split ovate concretion from the Spafford Shale, given the high abundance of calcitic brachiopod valves.

The contact between the Spafford Shale and Tichenor Limestone is a disconformity, a type of unconformity signifying a period of non-deposition or erosion due to a change in lithology (i.e.,



transition from shale to limestone). Furthermore, the Tichenor Limestone, at about 1m thick and the only carbonate unit present at Locality 1a, indicates an abrupt change in environmental setting after the deposition of the Spafford Shale. Limestones are typically deposited in warmer climates with low clastic input, so it is probable that the Tichenor Limestone formed in a warm, shallow sea (similar to the depositional environment of the Bahamas or Great Barrier Reef) (Fernandez, 2022). The presence of the tabulate coral *Favosites hamiltoniae* in an equivalent unit of the Tichenor Limestone at Portland Point (about two miles southeast of Salmon Creek) demonstrates that the unit was likely deposited in a warm, shallow reef system during the Givetian.



Inset right (above): a tabulate coral, *Favosites hamiltoniae*, present in an equivalent section of the Tichenor Limestone at Portland Point. Coral fossils provide further evidence that the Tichenor Limestone was deposited in a warm, well-oxygenated shallow marine setting with low clastic input.



Just above the Tichenor Limestone lies the Moscow Formation. Float from an undetermined section of the Moscow can be accessed at Locality 1a in talus piles that rest on the Spafford Shale platforms along the western bank of Salmon Creek (42°32'29.4"N 76°32'35.5"W). Fossils include spiriferid brachiopods, loxonematoid gastropods (*Palaeozygopleura* spp.), orthocone nautiloids (*Michelinoceras telamon*, *Spyroceras nuntium*), and phacopid trilobites. *Greenops* is the more common genus of phacopid than *Eldredgeops* in this unit. Well preserved *Zoophycos* (right) are present in calcareous silty mudstones and siltstones. Note that accessing the Lower Moscow float at Locality 1a is dependent on periodic landslides, low stage, and slow stream flow.



Inset right (previous page): although the seafloor was deeper during the time of deposition of the Moscow than the Tichenor Limestone, it remained sufficiently oxygenated to support infaunal (i.e., burrowing) organisms, evidenced by the presence of the trace fossil *Zoophycos*.



Fossils of the Lower Moscow Formation, accessed via a talus pile at Locality 1a (42°32'29.4"N 76°32'35.5"W). Left: mold of the phacopid trilobite *Greenops boothi*. Complete trilobite specimens such as this are rare (disarticulated pygidia are more typical). Right: a partial phragmocone of the orthocone nautiloid *Syproceras nuntium*.



Locality 1b. Ridge Road

Kashong Shale and Windom Shale (*Ambocoelia* Beds)

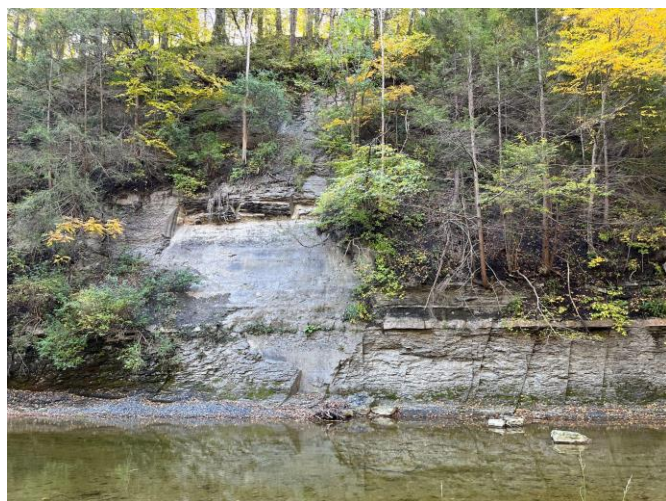
42°32'38.0"N 76°32'24.5"W



The Givetian strata exposed at Salmon Creek gently dips to the northeast. This can best be observed by following the height of the Tichenor Limestone relative to water level. Moving upstream to Locality 1b, the older Ludlowville Formation disappears below Salmon Creek; in its place are the younger Kashong and Windom Shale Members of the Moscow Formation, capped by the Tully Limestone (note that the Tully is not present at Locality 1).

the height of the Tichenor Limestone relative to water level from the right (downstream) to the Ridge Road Bridge (upstream). At Locality 1a, the Tichenor Limestone is approximately 5m above water level. At Locality 1b, the base of the Tichenor Limestone it is at water level. Accessing Locality 1b in fact requires climbing over the Tichenor.

In general, the Moscow Formation is composed of gray, fossiliferous shale with silty concretionary horizons. Both the Kashong and Windom Shale are net transgressive sequences in which water depth increased as a result of eustatic sea level rise (Brett et al., 2012). Therefore, both units are representative of deep marine environments.



About halfway between Localities 1a and 1b (42°32'33.7"N 76°32'27.5"W), the remains of a large rockfall are present on the southeastern bank of Salmon Creek. Collapse along a plane of weakness, likely a joint, brought down an approximate 10m section of Spafford Shale, Tichenor Limestone, and Lower Moscow. Large blocks of Tichenor Limestone are still present both on the Spafford Shale platforms along the southeastern bank as well as in the Creek bed itself. A collection of bluish-gray mudstones and light gray silty shales containing a similar faunal assemblage to the talus pile at Locality 1a



can be accessed from the Lower Moscow as float. Again, accessing this float is dependent on periodic landslides, low stage, and slow stream flow.

The Kashong Shale can be accessed just before reaching Locality 1b (42°32'37.0"N 76°32'24.6"W). It lies unconformably above the Tichenor Limestone, representing a change in depositional environment from a warm, shallow sea to a deeper marine basin. The Kashong Shale, a unit composed of a silty dark gray shale, remains to be thoroughly sampled at Salmon Creek for fossils. It appears to lack a diverse fossil assemblage, apparently containing mostly spiriferid brachiopods. However, the Kashong is noteworthy for its three-dimensionally preserved, tabulate coral *Thamnoptychia limbata* (inset right), unseen anywhere else in Salmon Creek.





Locality 1b, Ridge Road, is located on the southeastern bank of Salmon Creek. The Kashong Shale can be accessed above the Tichenor Limestone as it gently dips below water level ($42^{\circ}32'37.0''\text{N}$ $76^{\circ}32'24.6''\text{W}$). The *Ambocoelia* Beds of the Windom Shale is readily exposed at the slope beneath the Ridge Road Bridge.



The Windom Shale, in contrast to the Kashong, contains an abundance of fossils, with over 60 marine species described (Sang et al., 2013). At Locality 1b, fossils can be readily found in the *Ambocoelia* Beds, outcrops of fragile dark gray shale hosting an exuberant amount of the small spiriferid brachiopod *Ambocoelia umbonata*, for which the Bed gets its name. In addition to *A. umbonata*, other species of spiriferids (*Athyris spiriferoides*, *Mucrospirifer mucronatus*) as well as orthids (*Rhipidomella leucosia*) are present. Three subclasses of bivalves (Anomalodesmata, Palaeoheterodonta, and Paleotaxodonta) are represented in the *Ambocoelia* Beds, with *Nuculoidea corbuliformis* the most common. Compressed, partial phragmocones of the orthocone nautiloid *Spyroceras nuntium* are typical. Corals and crinoids are uncommon.



Inset left: common fossils from Locality 1b include the spiriferid brachiopods *Athyris spiriferoides* and *Mucrospirifer mucronatus* as well as thoracic segments and pygidia of the phacopid trilobite *Eldredgeops rana*.



The *Ambocoeila* Beds of the Windom Shale at Locality 1b is ideal for surface collecting.

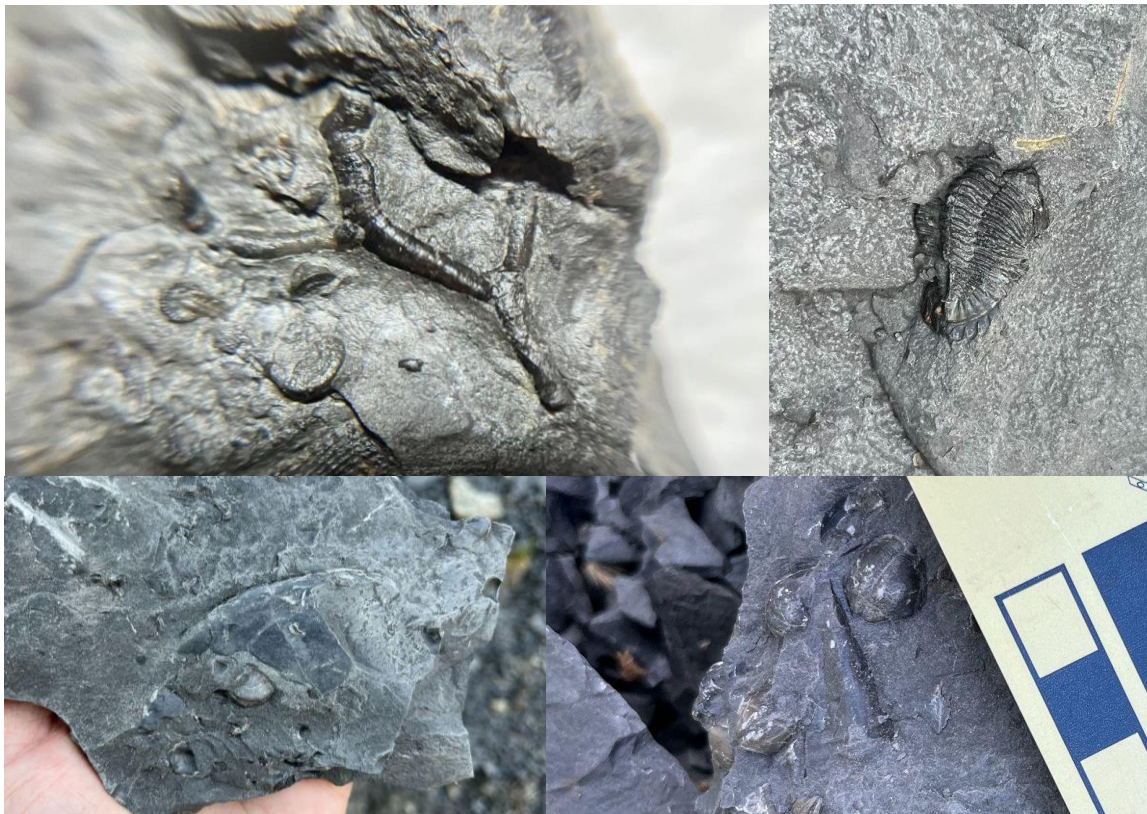
In addition, the *Ambocoeila* Beds boasts a bounty of phacopid trilobites, including *Eldredgeops rana* and occasionally *Greenops* (both *G. grabau* and *G. barberi*). Disarticulated molts compose the majority of phacopid fossils; however, complete specimens have been found.



Left: phacopid trilobite fossils at Locality 1b are generally disarticulated, with the cephalon (head) separated from the thorax and pygidium (body and tail, respectively). These fossils represent molts, or the shed exoskeletons of



trilobites. Right: complete, prone specimens such as this *Eldredgeops rana* are rare to come by. Even when complete trilobites are found, the fragility of the *Ambocoelia* Beds does not guarantee it will survive extraction (or the trip home) in one piece.



Uncommon fossil finds at Locality 1b. Top left: colony of the branching tabulate coral *Aulocystis dichotoma*. Top right: complete specimen of *Greenops grabau* (note the presence of the genal spine and occipital ring receding into the matrix, indicating a complete individual rather than a molt). Bottom left: partial carapace valve of the phyllocarid *Rhinocaris columbina*. Bottom right: calcareous pyramid shell of the hyolithid *Hallotheca aelis*, an extinct taxon of uncertain broader relationship to the known phyla, though possibly related to mollusks. Problematic taxa such as hyolithids, tentaculids, conulariids, and *Plumalina* are collectively referred to as “problematica” (Wilson, 2014).



Locality 2. Middle Salmon Creek (Ludlowville Road)

Windom Shale (*Zoophycos* churned beds, Bay View Coral Bed) and Tully Limestone

42°33'00.7"N 76°32'20.8"W (locality) | 42°33'02.2"N 76°32'17.7"W (parking)



Exposures of Windom Shale at Locality 2, facing downstream. The Tully Limestone is located several meters above the Windom Shale, out of view and behind foliage. Note the presence of large rectangular blocks of Windom Shale just above water level. These platforms, just like the ones present at Locality 1a, were created as a result of strong currents cutting into planes of weakness within the Shale during high stream stage (i.e., flooding events).

Moving upstream, the Givetian strata exposed at Salmon Creek continues its gentle downward plunge to the northeast. At Locality 2, neither the Tichenor Limestone nor the Kashong Shale are visible at the surface; in their place is the Windom Shale, capped by the Tully Limestone several meters above (note that the Tully is not accessible at Locality 2). The following will address several beds of interest within the Windom Shale at Locality 2, beginning at water level and working upwards in section. Refer to Figure 4 below.

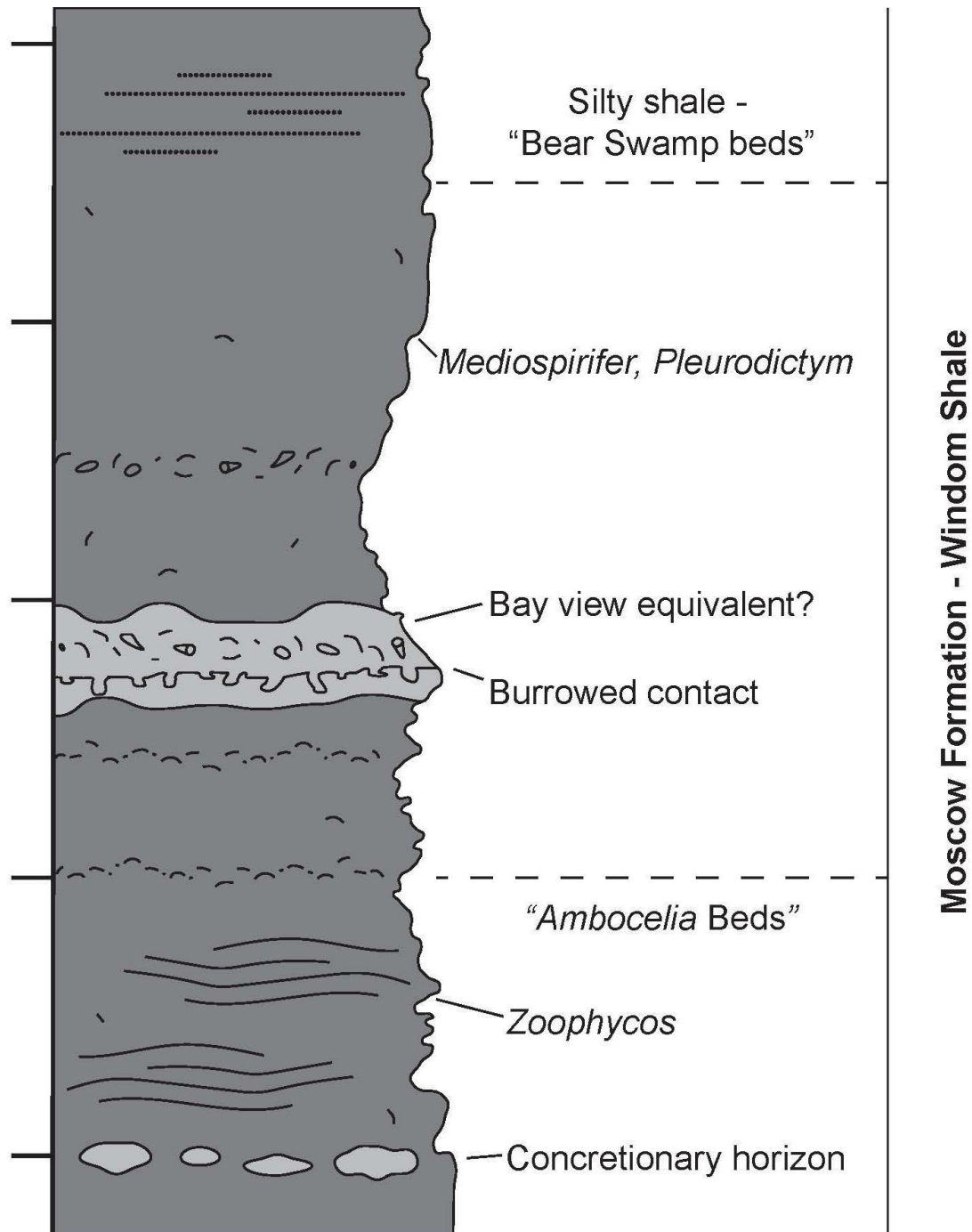


Figure 5. Measured section of the Windom Shale exposed at Locality 2, equivalent with exposed sections at Localities 3a and 3b (Hensen & Carnevale, 2023). Scale in meters.



The concretionary horizon of the Windom Shale at Locality 2 is generally inaccessible, as it is typically underwater (see Locality 3a). Just above this bed is a 1.5m thick series of light gray silty shale packages reworked by *Zoophycos* and thus referred to as the *Zoophycos* churned beds. *Zoophycos* is a trace fossil representative of sufficient oxygen conditions to support infaunal life in the sediment. In addition to *Zoophycos*, body fossils can be readily found weathering out of the rectangular platforms of Windom Shale at the base of Salmon Creek, accessible during low stage. Pavements of *Athyris spiriferoides*, similar to what can be seen in the Spafford Shale at Locality 1a, are typical. *Mediospirifer audaculus* is another common spiriferid brachiopod species found in these platforms. *Spinocyrtia granulosa*, though less common, is typically found with both valves preserved. Strophomenid brachiopods are represented by *Devonochonetes coronatus* and, in rare instances, *Megastrophia concava*. The orthocone nautiloid *Spyroceras nuntium* is relatively common in certain beds. Sometimes, long stalks of crinoids can be found sandwiched between individual beds of *Zoophycos* churned.



A spiriferid brachiopod pavement within the *Zoophycos* churned beds, dominated by *Athyris spiriferoides*. Additional species present include *Mucrospirifer mucronatus* and *Mediospirifer audaculus*.



Brachiopod fossils found *in situ* within the *Zoophycos* churned beds. Left: two valves of the spiriferid brachiopod *Mediospirifer audaculus*. Right: valve of the large strophomenid brachiopod, *Megastrophia concava*.



A long crinoid stalk within the *Zoophycos* churned beds.



A rare vascular plant, potentially a primitive lycophyte, *in situ* within the *Zoophycos* churned beds. This fossil represents some of the earliest plants to form coastal swamps during the Devonian Period. Situated at the boundary between the terrestrial and marine realms, pieces of these plants would occasionally be washed out to sea and ultimately deposited in the Catskill Delta and preserved as fossils.



Above the *Zoophycos* churned beds lies the Bay View Coral Bed. This Bed, as the name suggests, is dominated by assemblages of both rugose and tabulate corals; examples include *Streptelasma ungula* and *Pleurodictyum americanum*, respectively (Fig. 5). Ossicles and partial stalks of crinoids are also common. Platyceratoid gastropods (e.g., *Naticonema lineata*) occur but are rare. The presence of in-filled burrows within the Coral Bed emphasizes that the marine environment in which these animals lived was well-oxygenated, providing further evidence that the Windom Shale must have been deposited in a well-oxygenated setting in a shallow sea (e.g., the continental shelf).

Inset left: the Bay View Coral Bed. Note the presence of solitary rugose corals and disarticulated crinoid ossicles. This Bed is likely highly time-averaged.



Sections of the Upper Windom Shale above the Bay View Coral Bed can be accessed via talus piles that periodically accumulate as a result of failure along a plane of weakness (e.g., a joint), which commonly occurs during heavy precipitation.

For example, in October 2022, a silty section of the Upper Windom was brought down. The float contained an exceptionally preserved, large colony of the tabulate coral *Favosites hamiltoniae* (inset left). Unfortunately, this silty unit has since been swept away by Salmon Creek.



Additional fossils sampled from the silty shale unit. Left: a partial phragmocone of the orthocone nautiloid *Syproceras nuntium*. Top right: cast of a complete, prone specimen of the phacopid trilobite *Greenops grabau*.



Another rockfall occurred in April 2023, where a large block of calcareous siltstone from the upper Windom Shale was found as float along the northwestern bank of Salmon Creek (42°33'01.3"N 76°32'20.6"W). The following June, a period of torrential rainfall brought down additional pebbles of calcareous siltstone. The siltstones from both rockfall events are likely from the same unit in the Upper Windom Shale because they contain a

similar, highly diverse and highly abundant fossil assemblage. Calcified crinoid ossicles compose the majority of fossil material in most blocks from this unit. Fenestrid bryozoans are common. Calcified valves of spiriferid and orthid brachiopods are very common, including species such as *Mucrospirifer mucronatus* and *Tropidoleptus carinatus*, respectively. Interestingly, large, calcified valves of the strophomenid brachiopod *Mesoleptostrophia junia*



(inset left, previous page) can be found in relative abundance. Pleurotomarioid gastropods and orthocone nautiloids are rare.

Phacopid trilobites can also be found in relative abundance. In contrast to locality 1b, disarticulated pygidia of *Greenops* (both *G. grabau* and *G. barberi*) are quite common (*Eldredgeops rana* fossils have not been documented). In addition, thoracic segments of one of the largest trilobites of the Hamilton Group, *Dipleura dekayi*, are present.



Two examples of phacopid trilobite fossils collected as float from the calcareous siltstone talus pile of June 2023. Left: pygidium of *Greenops barberi*. Note the long, narrow, and triangular terminal lappet and pleural lappets with curved margins, contrasting it from *G. grabau*. Right: thoracic segment of *Dipleura dekayi*, identified from its large size and presence of pores on the surface (Wilson, 2014).



Inset left: a problematic fossil collected as float from the calcareous siltstone talus pile of June 2023. Consultations with several paleontologists at Carleton University, Cornell University, and Drexel University confirm that this fossil is a fish bone of either an acanthodian or placoderm. Other thoughts about the identification of this fossil include (1) trilobite caudal spine, (2) phyllocarid tail spike, or (3) eurypterid telson, though these are unlikely because (1) no phacopid trilobites from Salmon Creek possess caudal spines, (2) the fossil is too large to be a phyllocarid tail spike, and (3) eurypterids do not occur in the Windom Shale. Fish, however, have been found in the Windom Shale at Penn Dixie Quarry (Erie County, New York). Thus, this is likely the first reported fish fossil from the Windom Shale of Salmon Creek.



Uncommon fossil finds at Locality 2 from an undetermined bed of the Windom Shale (collected as float). Left: calyx of the camerate crinoid *Megistocrinus depressus*. Right: two carapace valves of the phyllocarid *Rhinocaris columbina*.

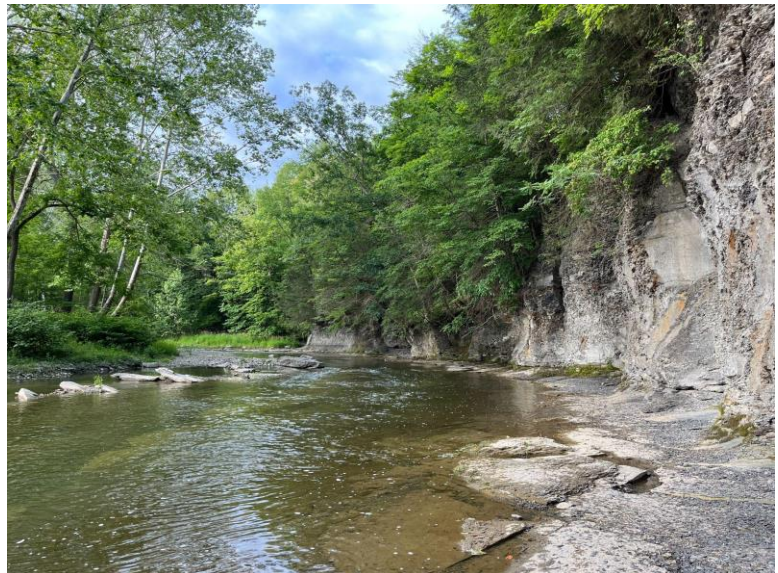
Locality 3. Upper Salmon Creek

Windom Shale and Tully Limestone
42°33'08.5"N 76°32'04.0"W

Locality 3a. Green Road

Windom Shale (concretionary horizon)
42°33'11.2"N 76°31'59.6"W

At Locality 3a, the concretionary horizon of the Windom Shale can be accessed during low stage (Fig. 5). This unit has not been heavily sampled. It appears to lack a diverse fossil assemblage, similar to the ovate concretions within the Spafford Shale at Locality 1a. However, occasional fossils of the phacopid trilobite *Eldredgeops rana* have been found weathering out of the concretions. These *E. rana* specimens are noteworthy because they are among the largest representatives of this species found in Salmon Creek.





Left: thoracic segments of the phacopid trilobite *Eldredgeops rana* weathering out of the concretionary horizon. Right: a relatively complete, prone specimen of *E. rana* from the concretionary horizon.

Locality 3b. Dug Road

Windom Shale (Bear Swamp Beds)

42°33'13.4"N 76°32'03.1"W

Locality 3b is the only known location in Salmon Creek between Myers Point and Ludlowville Falls where the Bear Swamp Beds of the Windom Shale can be accessed. This unit is composed of a series of thin shell beds dominated by spiriferid brachiopods. It is a complex sedimentary package of gray, silty mudstone. Large, concretionary burrows are present that cut across shell beds (inset left), indicative of reworked sediments and a well-oxygenated environment.

Spiriferids are represented commonly by *Mediospirifer audaculus* and uncommonly by

Spinocyrtia granulosa.

Both species can be found preserved either three-dimensionally or in “butterfly” (i.e., valves spread open). Fairly complete phragmocones of the orthocone nautiloid *Spyroceras nuntium* occur in relative abundance; they are typically compressed, but, interestingly, one was found retaining original biological shape. Large, disarticulated tagmata of the phacopid trilobite *Eldredgeops rana* are rare.





Uncommon finds at Locality 3b. From left to right: branchial valve of the spiriferid brachiopod *Spinocyrtia granulosa* (a three-dimensional specimen with both valves was extracted); phragmocone of the orthocone nautiloid *Spyroceras nuntium* retaining original biologic shape; complete molt of the phacopid trilobite *Eldredgeops rana* (note the cephalon was found disarticulated from the rest of the body).

Locality 3c. Ludlowville Falls

Windom Shale (*Ambocoelia* Beds) and Tully Limestone

42°33'16.8"N 76°32'11.8"W (locality) | 42°33'13.3"N 76°32'13.7"W (parking)



The Tully Limestone overlies the Windom Shale. Like the Spafford Shale-Tichenor Limestone contact at Locality 1a, the Windom Shale-Tully Limestone contact at Ludlowville Falls is a disconformity. Within the Tully itself is a series of smaller disconformities, represented at contacts



between the Carpenter Falls, Moravia, and Taughannock Falls members. Altogether, the Tully represents 500,000 years of time (Sang et al., 2013). Additionally, the Tully Limestone, a carbonate unit, indicates a change in environmental setting after the deposition of the Windom Shale. Just like the Tichenor Limestone at Locality 1a, it is probable that the Tully Limestone formed in a warm, shallow sea similar to the modern Bahamas or Great Barrier Reef.

At Locality 3c, Salmon Creek plunges about 50 feet over a fully exposed section of Tully Limestone, carving out the less resistant Windom Shale below (because shale is much softer than limestone, the Windom Shale weathers due to freeze-thaw and wetting/drying at a faster rate than the Tully Limestone, producing what is known as a caprock waterfall). Depending on the amount of rainfall, outcrops of the Windom Shale along the northeastern bank of Salmon Creek are accessible via a short trail from Ludlowville Park (or by wading upstream from Locality 3b).



The *Ambocoelia* Beds of the Windom Shale are exposed at Locality 3c (42°33'16.8"N 76°32'11.8"W). The Bed is best accessed along the northeastern bank of Salmon Creek, just downstream of Ludlowville Falls. Similar to Locality 1b, this section is composed of a fragile, dark gray shale that contains a plethora of the spiriferid brachiopod *Ambocoelia umbonata*, for which the Bed gets its name. In addition to *A. umbonata*, the spiriferids *Athyris spiriferoides* and *Mediospirifer audaculus* are very common. Orthid brachiopods, such as *Rhipidomella penelope* are less common. Unlike Locality 1b, the *Ambocoelia* Beds exposed at Locality 3c contain an abundance of corals, and are therefore more similar to Locality 2 in this regard. Rugose corals, such as *Streptelasma ungula* and *Stewartophyllum intermittens*, are more common than tabulate corals, including *Pleurodictyum*

americanum. Rugose corals at this locality are often calcitic. Phacopid trilobites are represented by *Eldredgeops rana*. Disarticulated tagmata, especially pygidia, are relatively common; complete specimens are rare. Unique to Locality 3c is the presence of globular pyrite nodules, which can often be found encrusted to the fossils. When exposed to the surface, these nodules rust in the presence of oxygen, changing color from yellow to reddish-brown.

Inset left: outcrop of the *Ambocoelia* Beds at Locality 3c. Three species of spiriferid brachiopod are present, ordered from most to least abundant: *Ambocoelia umbonata*, *Mediospirifer audaculus*, and *Athyris spiriferoides*. Note the reddish-brown nodules of pyrite encrusted on many of the spiriferids.

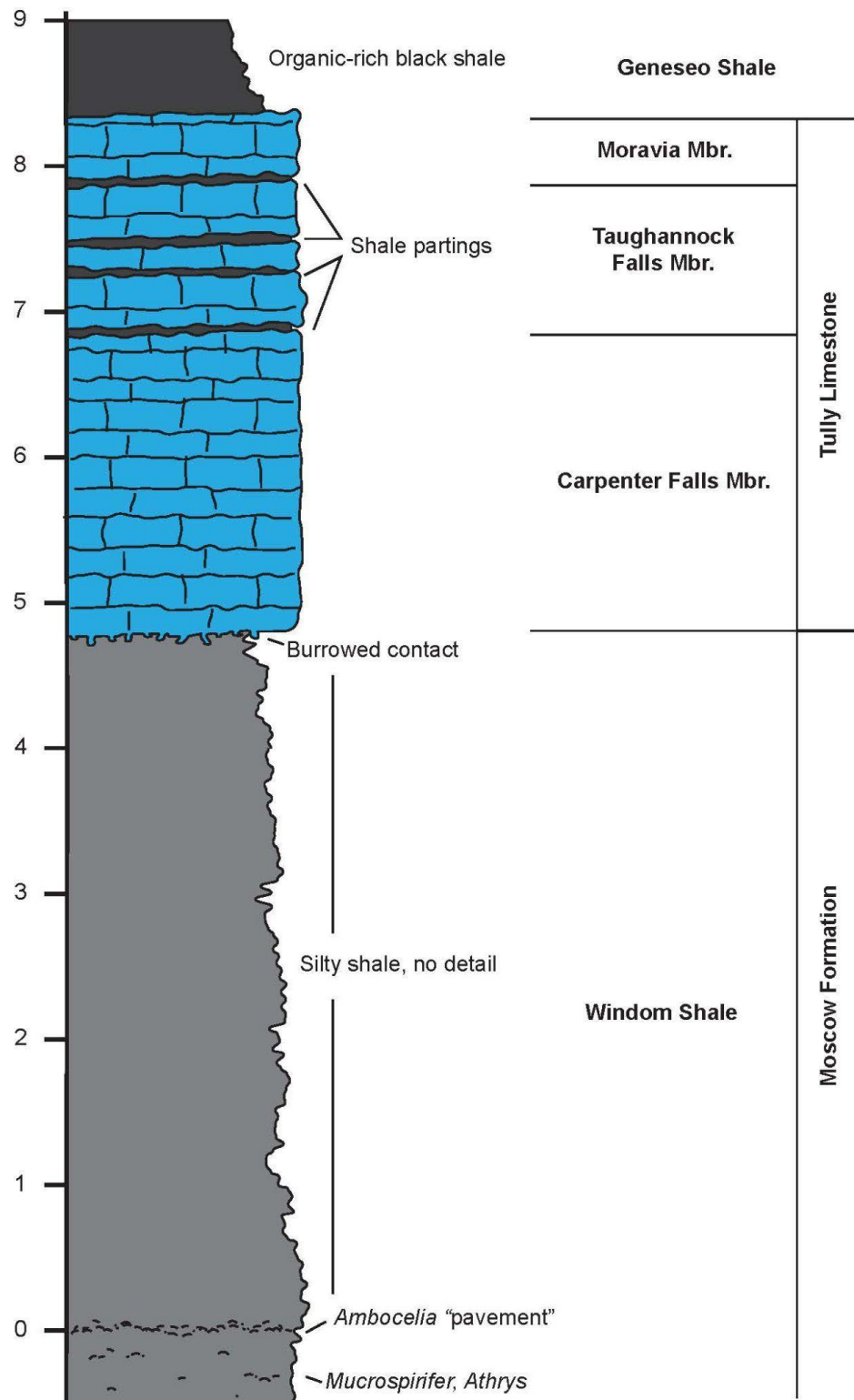


Figure 6. Measured section of the Moscow-Tully interval exposed at Locality 3. The Genesee Shale can be seen upstream of the falls, in exposures along Salmon Creek Road. Scale in meters.



Mollusks appear to be very rare at Locality 3c. However, a complete, pyritized *Tornoceras uniangulare* was shown to the primary author for identification during Fossil ID Days at the Paleontological Research Institute in June 2023. The fossil was said to have come from Ludlowville Falls. The presence of pyrite indicates it likely came from the *Ambocoeila Beds*. This fossil is the first goniatite ammonoid to be reported from Salmon Creek.



Common finds at Locality 3c. Left: valves of the spiriferid brachiopod *Mediospirifer audaculus*. Right: calcitic corallum of the stauriid rugose coral *Streptelasma ungula*.



A complete, prone specimen of *Eldredgeops rana* with pyrite nodular growth (i.e., “pyrite disease”) is a rare find at Locality 3c.

During low stage in the summer, the underside of Ludlowville Falls is accessible (42°33'17.2"N 76°32'14.5"W). Large locomotive burrows, potentially made by trilobites, can be seen looking up at the base of the Tully Limestone, indicating a well-oxygenated ocean bottom supported a diversity of marine faunas during the time of deposition.



Large burrows seen beneath Ludlowville Falls within the base of the Tully Limestone.



Comparison of the stage level of Salmon Creek and Ludlowville Falls during a dry period in October 2022 (left) and after heavy rain in November 2022 (right).

The Tichenor Limestone and Tully Limestone can be compared at Localities 1a and 3c. The Tichenor Limestone of the Ludlowville Formation is exposed at Locality 1a and about 1m thick, though it may be up to two meters thick at other localities. It is probable that the components of the Tichenor Limestone were not bound together during deposition. Overall, the Tichenor Limestone lacks mud and supports grains and is thus considered to be a grainstone. The Tully Limestone exposed at Locality 3c is about nine meters thick. Differences in siliciclastic content allow the Tully Limestone to be subdivided into Lower, Middle, and Upper sections, composed of shales, siltstones, and sandstones, respectively (Wilson, 2014). The Tully Limestone supports mud and includes more than ten percent grains. It can therefore be classified as a wackestone.





Comparison of the Tichenor Limestone (left) with the Tully Limestone (right) from equivalent units at Portland Point, Lansing, NY and Taughannock Falls State Park, Trumansburg, NY, respectively. Notice the presence of grains and absence of mud in the Tichenor Limestone (a grainstone) and absence of grains and presence of mud in the Tully Limestone (a wackestone). In addition, the tabulate coral *Favosites hamiltoniae* and a network of burrows are visible in the Tichenor and Tully Limestone, respectively.



Geologic History of Salmon Creek and Ludlowville Falls after the Givetian Stage

The last unit of strata exposed at Salmon Creek is the Tully Limestone, present at both Localities 2 and 3c. The Tully Limestone marks the end of the Givetian-aged Hamilton Group. Units younger than the Tully Limestone, such as the Frasnian-aged (early Late Devonian) Ithaca Formation of the Genesee Group, can be seen on Cornell University campus at Fall Creek, Beebe Lake, and Cascadilla Gorge. A full sequence from the Tully Limestone to Ithaca Formation is exposed at the Upper Falls of Taughannock Falls State Park in Trumansburg, NY.

The Ithaca Formation is one of the last Devonian units exposed in western and central New York State. By the end of the Frasnian, the Catskill Delta progradation completed and the



Appalachian Basin was entirely filled with sediment. As a result, the Devonian marine record in New York State ended. From the Frasnian to the Late Pleistocene, the rate of erosion generally outpaced the rate of deposition. Consequently, an enormous unconformity exists between the Devonian strata present at Salmon Creek (deposited around 385 Ma)

and the overlying glacial till deposited during the Late Pleistocene Epoch (11.7 Ka). An exception to this pattern occurs only in select localities, such as in Six Mile Creek (Ithaca, NY), where Jurassic- and Cretaceous-aged kimberlite (inset left) can be found intruded between the Frasnian-aged shales of the Ithaca Formation.

Following the Late Pleistocene, New York State experienced several successions of glacial advance and retreat. Glacial advance reached an acme around 21 Ka and final retreat began around 11 Ka. The Finger Lakes of central New York State originated as north-south river valleys. Glacial retreat ultimately deepened these pre-existing river valleys (Sang et al., 2013). Deposition of glacial till during episodes of retreat impounded meltwater, forming proglacial lakes. As glacial retreat continued, the proglacial lakes slowly drained and fell in elevation, carving steep inclines into the landscape. These steep inclines increased the distance from the base of the proglacial lakes to the height at which a stream, such as Salmon Creek, fed into it. Water follows the path of least resistance, so over time, Salmon Creek carved through sediment along its new path toward the present-day Cayuga Lake, exposing Givetian strata beneath it. Salmon Creek cut into soft rock, such as the Windom Shale, at a quicker rate than through harder rock, such as the Tully Limestone. Ludlowville Falls formed this way as the Creek plunged over harder units of caprock, cutting ever deeper into soft shales. Today, Salmon Creek deposits sediments as it meets Cayuga Lake, prograding to the southwest as it builds up Myers Point.



Appendix

The Fossils of Salmon Creek and Ludlowville Falls

Arthropoda

Trilobita

Trilobites are an extinct class of marine arthropods. All arthropods share jointed appendages, a segmented body, and a chitinous exoskeleton. For these reasons, trilobites are distant relatives of insects, spiders, horseshoe crabs, and crustaceans (shrimp, crabs, and lobsters) (Benton and Harper, 2020a).

Trilobites are composed of three basic parts: the (1) cephalon (head), (2) thorax (body segments), and (3) pygidium (tail). The body can also be divided into three lobes: two lateral pleural lobes (PL) and one medial axial lobe (AL). This is where trilobites get their name (“trilobite” means “three lobes”) (Wilson, 2014).

Exposures of Windom Shale at Salmon Creek contain trilobites belonging to the class Phacopida. *Eldredgeops rana*, formerly *Phacops rana*, is probably the most common trilobite from the Middle Devonian Hamilton Group. The name “rana”, in fact, means “frog”, referring to the frog-like eyes of this trilobite. *E. rana* occurs in abundance in the *Ambocoeilia* Bed (Locality 1b). *Greenops* is another relatively common genus at Salmon Creek, represented by *G. grabaui* and *G. barberi*. Both species can be collected as float (talus piles at Localities 1a and 2). It is probable that *Bellacartwrightia phyllocaudata* is also present at Salmon Creek. It has not been documented yet, likely owing to its similar appearance to *Greenops*. *Dipleura dekayi* is one of the largest phacopid species of the Hamilton Group and has been found in an undetermined bed of the Upper Windom Shale (talus pile at Locality 2)

The most common type of trilobite fossils are incomplete molts (i.e., shed exoskeletons), which can be found throughout the Windom Shale (Localities 1b, 2, 3a, 3b, and 3c). Complete trilobite specimens (prone or enrolled) are rare. Complete, prone specimens have been found in the *Ambocoeilia* Beds (Localities 1b and 3c) and concretionary horizon (Locality 3a).

Phyllocarida (“sea fleas”)

Phyllocarids are an extant subclass of shrimp-like crustaceans with a cephalothorax encased in a bivalved carapace and an abdomen that ends in a spiked tail. As crustaceans, they are closely related to the decapods, the clade that includes the crabs, lobsters, and “shrimp.”

Phyllocarids first appeared in the Cambrian and rapidly diversified during the Ordovician. They are still around today, having survived the big five mass extinctions. For this reason, the behavior of living phyllocarids can be used to infer the behaviors of fossil phyllocarids. Since extant phyllocarids use their antennae and/or legs to kick up the sediment and filter out suspended particles, it is probable that fossil phyllocarids did the same. Interestingly, some fossil phyllocarids have robust mandibles, unlike living forms; they may have used these structures for crushing hard-shelled prey (durophagy).



In western/central NY, four genera of fossil phyllocarids can be found, though they are rather uncommon. *Rhinocaris columbina* is one species that occurs throughout the Hamilton Group. It is diagnosed by an elongate valve with lessly running raised lines and dorsal pits. Typically, only one of the two valves is found, and it is often crushed. At Salmon Creek, partial carapace valves of *R. columbina* can be found in rare instances in the *Ambocoelia* Bed (Locality 1b) and as float likely from the Upper Windom Shale (Locality 2). Evidence of other species of phyllocarid, such as *Echinocaris punctata*, have not yet been documented at this locality, even though they co-occur with *R. columbina*. Mandibles, abdomen, or tail spikes have not yet been documented.

Brachiopoda (“lamp shells”)

Brachiopods, also known as lampshells, are a phylum of lophophorate invertebrate. Like bivalves, brachiopods have twin valves. Unlike bivalves, all lophophorates have a feeding structure called a lophophore. This makes brachiopods more closely related to bryozoans and phoronids than to bivalves. Brachiopods first appeared in the Cambrian and their peak diversity occurred during the Paleozoic Era (Benton and Harper, 2020b). Modern brachiopods live in cryptic, deep-water environments (this is why most people have never seen a living brachiopod).

The most abundant order of brachiopods found at Salmon Creek are spiriferids. Common spiriferids include *Ambocoelia umbonata*, *Athyris spiriferoides*, and *Mucrospirifer mucronatus*. Uncommon spiriferids include the larger species *Mediospirifer audaculus* and *Spinocyrtia granulosa*. *Megastrophia concave* and *Mesoleptostrophia junia* are two rare, large species of strophomenid. Relatively common orthids include *Tropidoleptus carinatus* and *Rhipidomella leucosia*.

Both bivalve and brachiopod fossils can be found at Salmon Creek. How do you tell the difference? The trick is symmetry: bivalves are symmetrical between valves whereas brachiopods are symmetrical across valves. In other words, a single bivalve shell is asymmetrical whereas a single brachiopod shell is symmetrical (Fig. 5).

**Brachiopod****Bivalve**

Figure 5. Symmetrical differences between brachiopods and bivalves. Adapted from Pier (n.d.).

Bryozoa (“moss animals”)

Bryozoans are a group of colonial invertebrates that resemble corals and hydroids. They are related to brachiopods and phoronids because they all share a complex feeding structure called a lophophore.

An individual bryozoan is known as a zooid, and each zooid is enclosed in a calcareous exoskeleton and attached to the colony by a stolon. Each zooid has a specialized function to serve the larger colony, such as feeding, defending, reproducing, or removing excess sediments. Bryozoan colonies come in many different shapes, from encrusting and sheet-like to branched forms (Benton and Harper, 2020b).

Bryozoans are the only major animal phylum absent during the Cambrian Explosion. They appeared later in the Early Ordovician (Allmon, 2022c). About 6,000 living species can be found today, typically attached to other organisms such as seaweeds in the intertidal zone.

Common bryozoans at Salmon Creek include the fan-shaped fenestrids, which can be found in the Spafford Shale (Locality 1a).

Cnidaria (jellies, corals, and allies)

Anthozoa (sea anemones, soft corals, stony corals)

Two orders of fossil corals can be found in central New York State, (1) rugose and (2) tabulate corals (Wilson, 2014). Both rugose and tabulate corals first appeared in the Ordovician. They went extinct during the end-Permian mass extinction event (Benton and Harper, 2020c).

Interestingly, after the disappearance of rugose and tabulate corals, there were no corals in the oceans for at least 14 million years until the appearance of scleractinian corals (Stanley, 2003).



Rugosa (“horn corals”)

Rugose corals include both solitary and colonial forms. Rugose corals were nestlers that sat in the mud on the ocean bottom. As they grew, they would fall over. Over time, this produces the curved horn structure of rugose corals (hence their common name, the horn corals) (Wilson, 2014). Common species of rugose corals at Salmon Creek include *Streptelasma ungula* and *Stewartophyllum intermittens* and can be found in the Bay View Coral Bed (Locality 2) and *Ambocoelia* Beds (Locality 3c).

Tabulata

Tabulate corals are entirely colonial. The shape of a colony varies from sheet-like to branching forms. It has been proposed that tabulate corals, like their scleractinian relatives, may have had photosymbionts (Allmon, 2022a). *Favosites hamiltoniae* commonly occurs in the Tully Limestone (e.g., Portland Point, Lansing, NY) and has been collected as talus from the Upper Windom Shale (Locality 2). *Pleurodictyum americanum* can be found in the Bay View Coral Bed (Locality 2) and *Ambocoelia* Beds (Locality 3c). *Aulocystis dichotoma* is an uncommon encrusting species associated with spiriferid brachiopods.

Echinodermata (sea stars, brittle stars, sea cucumbers, and allies) Crinoidea (“sea lilies”)

Although they look like plants, crinoids are animals related to sea stars, brittle stars, sea cucumbers, sea urchins, and sand dollars. They first appeared in the Early Ordovician and were abundant throughout the Paleozoic. Today they are less common, inhabiting cryptic habitats such as deep-water environments (Allmon, 2022b).

Crinoids have root-like holdfasts to attach themselves to the seafloor. The stalk, or stem, is made up of many individual components called ossicles, and the theca is composed of many calcitic plates. Several arms branch off from the calyx and aid with feeding (Wilson, 2014).

Crinoids are probably the second most abundant fossil in western and central New York State, after brachiopods. Like their sea star cousins, crinoids disarticulate (come apart) after death, so they are commonly found as individual ossicles, making identification to lower taxonomic ranks impossible.

Ossicles and partial stalks of crinoids can be commonly found in the *Zoophycos* churned beds and Bay View Coral Bed of the Windom Shale (Locality 2). *Megistocrinus depressus* is the only crinoid from Salmon Creek identified to the species rank due to discovery of a partial, yet diagnostic, calyx from an undetermined bed of the Windom Shale (Locality 2). Complete crinoid fossils are rare and are unlikely to occur at Salmon Creek given the environments of deposition preserved.

Ichnotaxa (trace fossils)

Swirly impressions on shales present at Salmon Creek are in fact evidence for feeding behavior about 380 million years old. These structures, known as *Zoophycos*, were probably produced by the sweeping motions of a proboscis of a worm or another burrowing creature to survey the greatest area for food while spending the least amount of energy when feeding in the sediment. Since many different kinds of organisms could have made these structures, *Zoophycos*, like all



trace fossils, is the name for this particular type of structure and not the maker of it (Allmon, 2022d). *Zoophycos* can be commonly found in the Lower Moscow (talus pile at Locality 1a) and *Zoophycos* churned beds (Locality 2).

Other common trace fossils include simple burrows. Small, in-filled burrows can be found reworking sediments in the Bay View Coral Bed (Locality 2). Similarly, medium-sized, concretionary burrows occur in the Bear Swamp Beds (Locality 3b), cutting across individual shell beds of spiriferid brachiopods. Very large burrows can be seen underneath Ludlowville Falls within the base of the Tully Limestone (Locality 3c), likely made by trilobites.

Incertae sedis (“problematica”)

Problematic taxa such as hyolithids, tentaculids, conularids, and *Phumalina* are collectively referred to as “problematica”, as they all are extinct taxa of uncertain broader relationships to the 36 currently recognized animal phyla (i.e., they are problems for taxonomists).

Hyolithida

Hyoliths are a problematic clade of lophophotrochozoans that first appeared in the Fortunian (Early Cambrian). They all share calcareous pyramidal shells and an unhinged operculum (“trapdoor”). Though unhinged, the operculum indicates a potential relationship to the mollusks (bivalves, gastropods, and cephalopods).

Two suborders of hyoliths are recognized: (1) Hyolithida and (2) Orthothecida, the former occurring in the Middle Devonian Hamilton Group of western and central NY State. Both hyolithids and orthothecids were benthic organisms, using tentacles to capture food. In addition to tentacles, orthothecids have paired appendages that were perhaps used to help these organisms position themselves like a tripod and survey food in the water column instead of within the sediment.

Only three individuals of one species of hyolithid, *Hallotheca aclis*, has thus far been found at Salmon Creek. Despite occurring throughout the entire Hamilton, all three have been discovered in only the Windom Shale Member of the Moscow Formation. Both *H. aclis* and *Hyolithes striatus* went extinct during the Kellwasser event at the end of the Givetian, and hyoliths disappeared entirely during the end-Permian mass extinction event.

Mollusca

Bivalvia (clams, scallops, oysters, mussels)

Bivalves (formerly known as pelycopods) are a class of marine and freshwater mollusks that include clams, oysters, mussels, and scallops. Most are suspension feeders. The group first appeared in the Cambrian Period and rapidly diversified in the Cenozoic Era. All bivalves have two-hinged shells (hence the name “bi-valve”) surrounding the stomach, a foot, and gills (Allmon and Mikkelsen, 2020).

Subclasses of bivalves found at Salmon Creek include Anomalodesmata (*Grammysioidea alveata*), Heterodonta (*Cypricardella bellistriata*), Palaeoheterodonta (*Modiomorpha* spp., *Goniophora hamiltonensis*), and Pteriomorphia (*Pseudoaviculopecten princeps*, *Cornellites fasciculata*). All can be found in the Spafford Shale (Locality 1a).



Cephalopoda (squids, octopuses, cuttlefishes)

Cephalopods are a class of active marine predatory mollusks that first appeared in the Cambrian. All cephalopods share a funnel, arms (tentacles) surrounding the mouth, and a chitinous beak (Hendricks, 2018).

Nautiloidea

Nautiloids are a subclass of cephalopods. Straight shelled nautiloids, called orthocones, are the most common cephalopod fossil from central NY, but there are many other forms of nautiloid. Though the orthocones are extinct, nautiloids as a group are not. The Chambered Nautilus (*Nautilus pompilius*), for example, is a living relative of the Givetian-aged orthocones at Salmon Creek.

Unlike true squids (subclass Coleoidea), nautiloids have external shells called phragmocones (coleoids have reduced and internalized this structure – referred to as the cuttlebone). The individual chambers of the shell are called septa, and nautiloids regulate the amount of gas in their septa to control their buoyancy in the water column, like a submarine (Hendricks, 2018).

Three species of orthocone nautiloids can be found in the Moscow at Salmon Creek: (1) *Spyroceras nuntium*, (2) *Dolorthoceras exile*, and (3) *Michelinoceras telamon*. *S. nuntium* is the most common.

Ammonoidea

Ammonoids are an extinct subclass of coiled cephalopods. They first appeared in the Devonian and disappeared during the end-Cretaceous mass extinction event. There are three groups of ammonoids, but only goniatites, those lacking subdivided saddles and lobes along their sutures, can be found in the Middle Devonian Hamilton Group of western and central New York State. Goniatites went extinct during the end-Permian mass extinction event. They were replaced by the ceratites in the Triassic, which in turn were replaced by the ammonites in the Jurassic (Hendricks, 2018). Only one goniatite ammonoid (*Tornoceras uniangulare*) has been reported from Salmon Creek (Locality 3c).

Gastropoda (snails, slugs)

Gastropods are a class of marine, freshwater, and terrestrial mollusks that first appeared in the Cambrian. The name “gastropod” comes from the Greek roots “gastros-” meaning “stomach” and “-pod” meaning foot.

All gastropods have a head, a single valve (shell), operculum (if shell is present) on the foot, and experience torsion during development. The shell has been extensively modified (in terms of elongation and coiling) by various subclasses of gastropods. Some gastropods, such as the nudibranchs, even lost their shells.

The radula (a tongue-like band covered by rows of chitinous, bristle-shaped teeth) is a specialized structure present in all gastropods, allowing them to have a variety of feeding types (herbivory, carnivory, detritivory, scavenging, and parasitism). Carnivorous gastropods use the radula to drill into shelled prey, including bivalves, brachiopods, and other gastropods, leaving behind drill holes called borings (Allmon and Mikkelsen, 2021).



Two clades of gastropods can be found at Salmon Creek: (1) Pleurotomarioidea (*Bembexia sulcomarginata*) and (2) Loxonematoidea (*Palaeozygolpeura* spp.). Pleurotomarioids are less common than loxonematoids.

Plantae

Tracheophyta (vascular plants)

Occasionally, plant material was transported from the terrestrial realm to the sea, where they would ultimately be deposited in the Catskill Delta Complex of the Appalachian Basin and preserved as fossils. A fossil of a potential lycophyte, a primitive vascular plant and component of the first coastal swamps, has been found in the *Zoophycos* churned beds of the Windom Shale at Salmon Creek (Locality 2).

Vertebrata

Acanthodians (“spiny sharks”) and other Devonian fishes

Miscellaneous plates and bones of fishes, including ostracoderms, placoderms, and primitive osteichthyans (bony fishes), have been found in the Windom Shale and North Evans Limestone at Penn Dixie Quarry in Erie County, NY. Typically, Penn Dixie fish fossils are so fragmentary that they cannot be identified to lower taxonomic ranks (Stokes and Schreiber, 2017). However, in one very rare case, a lower jaw, complete with teeth, was found just to the south of Penn Dixie at Eighteenmile Creek. It is believed to belong to an acanthodian, a basal cartilaginous fish that superficially resembles modern sharks (hence the common name, “spiny shark”).

Moving east towards Salmon Creek, the likelihood for encountering fish material in the Windom Shale dramatically decreases. On one occasion, a fish bone was discovered as float, probably from the Upper Windom Shale (talus pile at Locality 2). Consultations with paleontologists at Carleton University, Cornell University, and Drexel University indicate that the bone is most likely a fin spine from an acanthodian.



Summary Chart of the Distribution of Fossils at Salmon Creek and Ludlowville Falls

VC = Very Common, C = Common, U = Uncommon, R = Rare, VR = Very Rare

	Spafford Shale	Kashong Shale	Windom Shale: <i>Ambocoelia</i> Bed	Windom Shale: Concretionary Horizon	Windom Shale: <i>Zoophycos</i> churned	Windom Shale: Bay View Coral Bed	Windom Shale: Bear Swamp Bed	Locality 1 undetermined float (Lower Moscow)	Locality 2 undetermined float (Windom Shale)
Arthropoda:									
Phyllocarida:									
<i>Rhinocaris columbina</i> Clarke, 1888			VR			VR			
Trilobita:									
<i>Bellacartwrightia phyllocaudata</i> Lieberman & Kloc, 1997					VR				
<i>Dipleura dekayi</i> (Green, 1832)									VR
<i>Eldredgeops rana</i> Green, 1832	VR	C	R				R	VR	
<i>Greenops</i> spp.		R					R	R	U
Brachiopoda:									
Orthida:									
<i>Rhipidomella leucosia</i> (Hall, 1860)		U							C
Spiriferida:									
<i>Ambocoelia umbonata</i> (Conrad, 1842)	C	VC							
<i>Athyris spiriferoides</i> (Eaton, 1831)	VC	C		VC					
<i>Mediospirifer audaculus</i> (Conrad, 1842)				C		C			
<i>Mucrospirifer mucronatus</i> (Conrad, 1841)	C	C		C					VC
<i>Spinocyrtia granulosa</i> (Conrad, 1839)				U		U			VR
Strophomenida:									
<i>Devonochonetes coronatus</i> (Conrad, 1842)				R					R
<i>Megastrophia concava</i> (Hall, 1857)				VR					
Bryozoa:									
<i>Stenolaemata</i> Borg, 1926		VR						R	
Cnidaria									
Rugosa:									
<i>Stewartophyllum intermittens</i> Busch, 1941			VC						
<i>Stereolasma rectum</i> Hall, 1876		U							
<i>Streptelasma ungula</i> Hall, 1876			VC			VC			
Tabulata:									
<i>Aulocystis dichotoma</i> (Grabau, 1899)			VR						

[illegible]



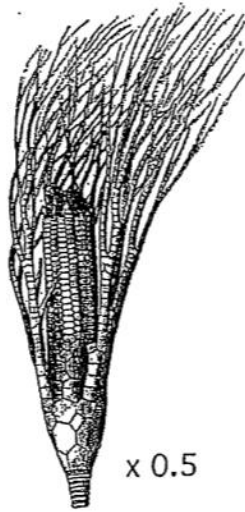
Echinodermata



crinoid stem pieces



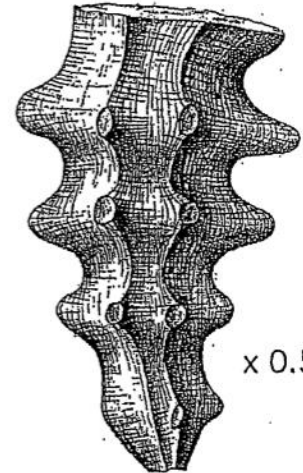
crinoid stem



x 0.5

crinoid calyx with arms

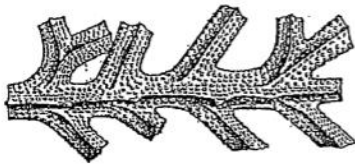
Porifera



x 0.5

Hydnoceras

Bryozoa

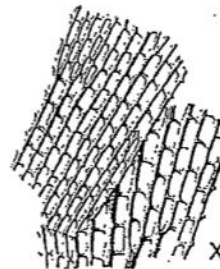


Taeniopora



x 2

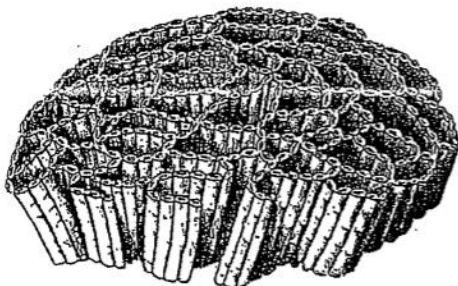
cyclostome



x 2

fenestrate

Anthozoa



Halysites



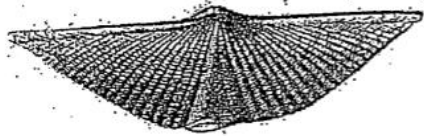
Heterophrentis



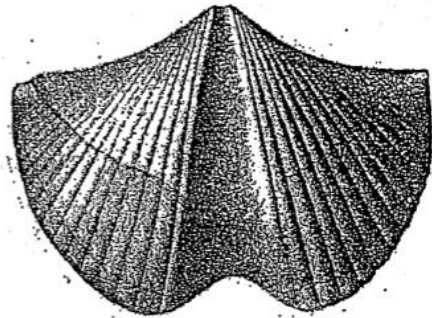
Heliophyllum halli



Brachiopoda



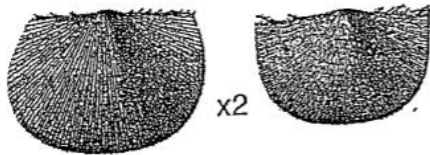
Mucrospirifer mucronatus



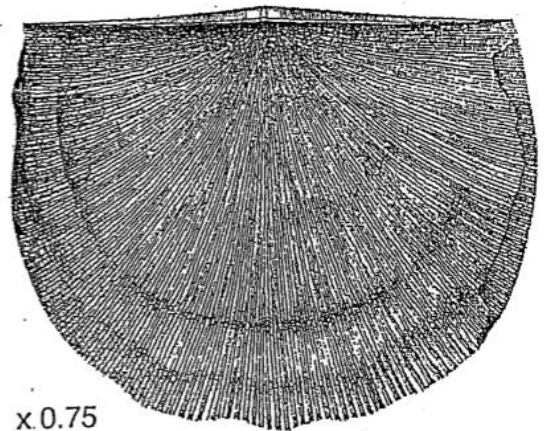
Spinocyrtia granulosa



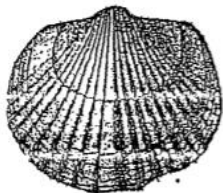
Athrys spiriferoides



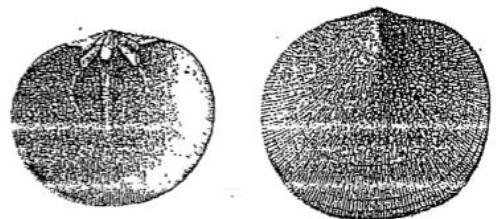
Devonochonetes coronatus



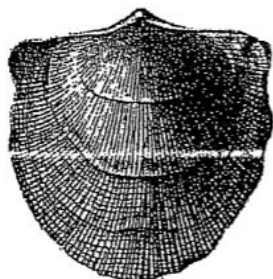
Megastrophia textilis



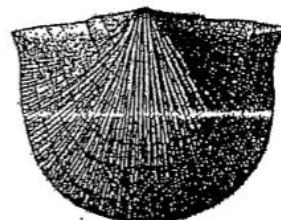
Tropidoleptus carinatus



Rhipidomella penelope



Pseudatrypa devoniana

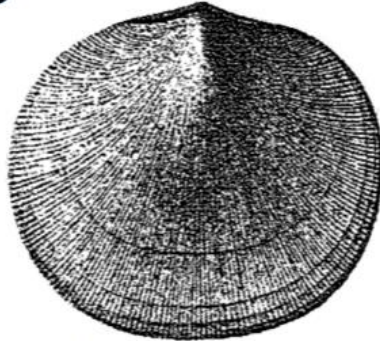
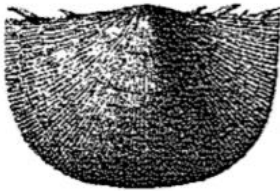


Strophodonta demissa



brachiopods

Devonchonetes coronatus

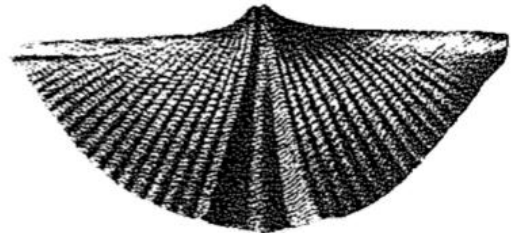
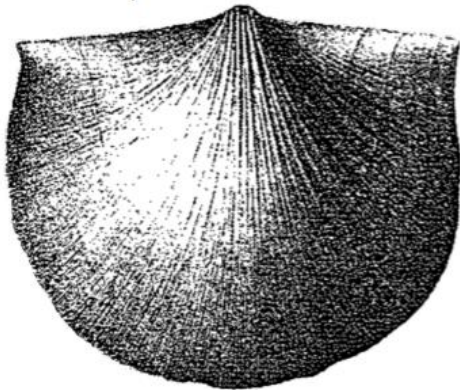


Rhipidomella penelope



Athyris spiriferoides

Strophodonta demissa



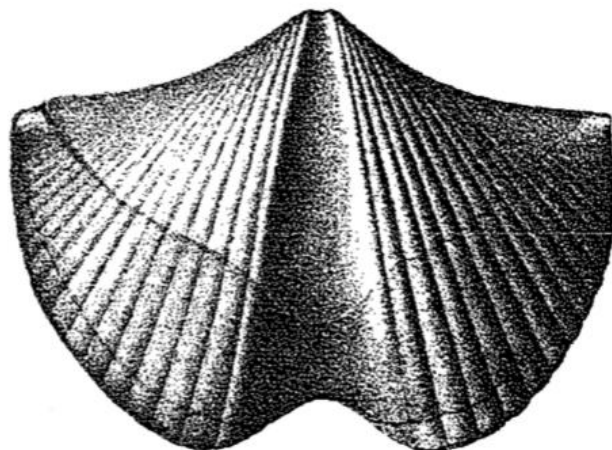
Mucrospirifer mucronatus



Ambocoelia umbonata



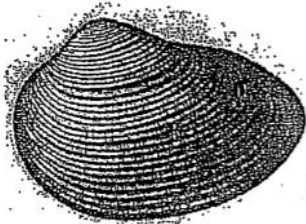
Tropideleptus carinatus



Spinocyrtia granulosa

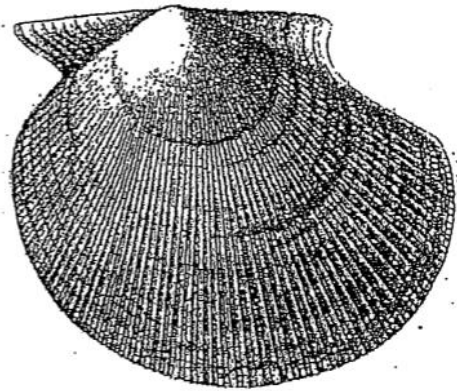


Bivalvia

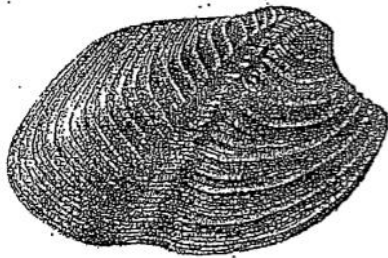


x2

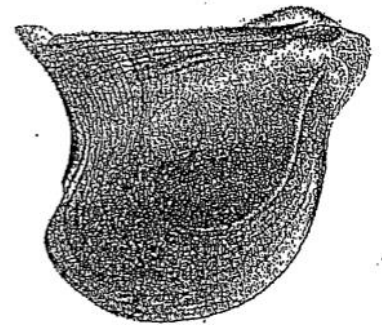
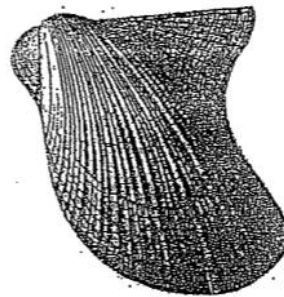
Nuculoidea



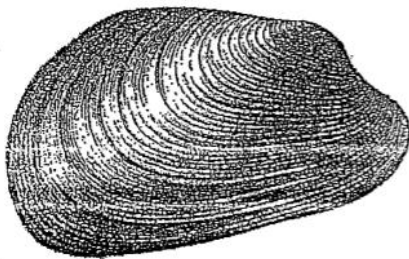
Pseudoaviculapecten princeps



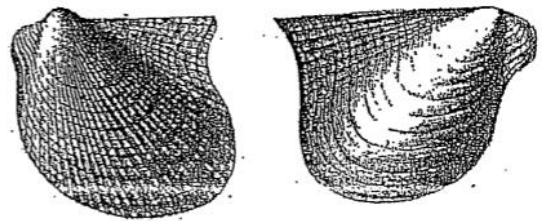
Grammysia bisulcata



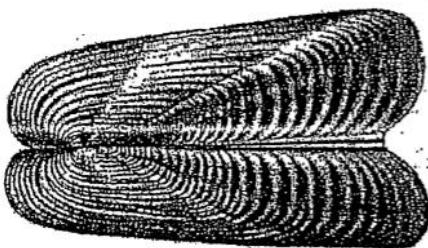
Cornellites fasciculata



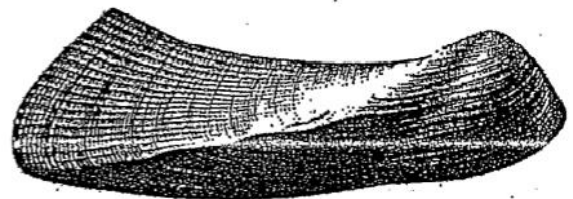
Modiomprha mytiloides



Actinopteria boydi



Orthonota undulata



Cimitiara recurva



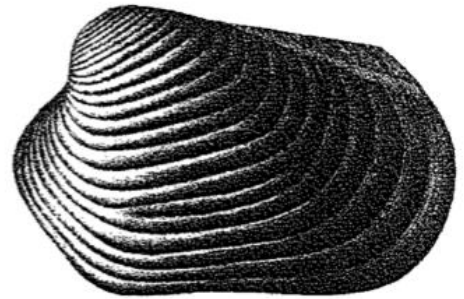
bivalves



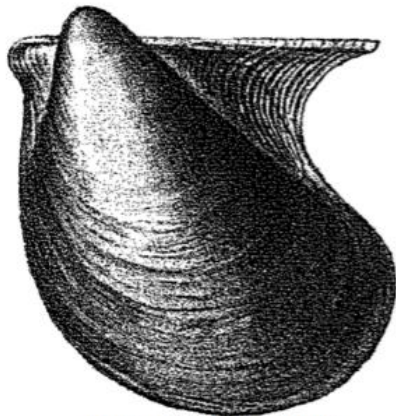
Nuculoidea lirata



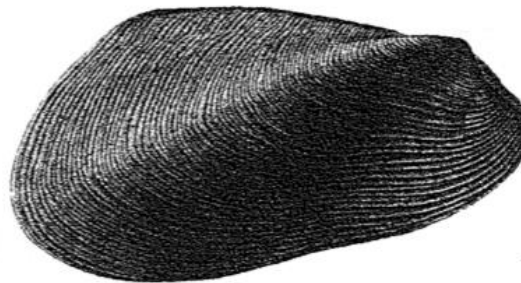
Cypricardella bellistriata



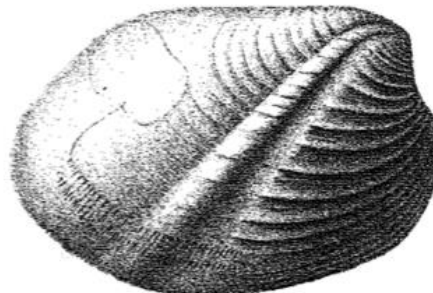
Grammysioidea arctuata



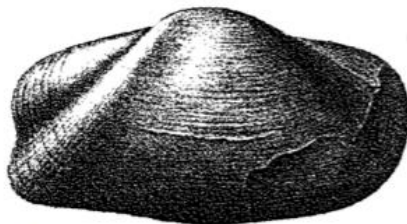
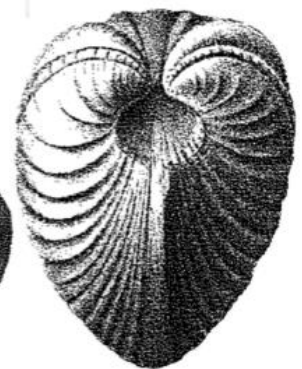
Leiopoteria conradi



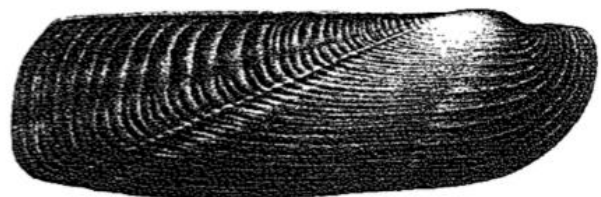
Modiomorpha concentrica



Grammysia bisculata



Telinopsis subemarginata



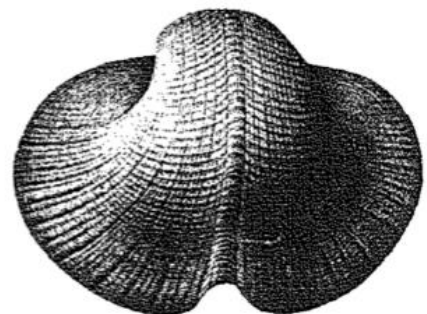
Orthonota undulata

gastropods

Palaeozygopleura hamiltoniae



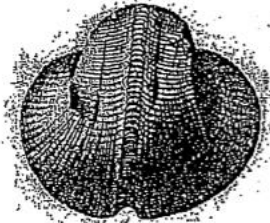
Reudemannia trilix



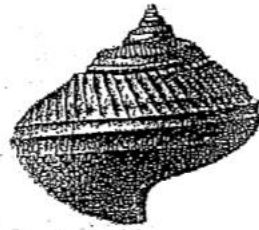
Retispira leda



Gastropoda



Bellerophon



Bembexia sulcomarginata



x 2

Paleozygopleura hamiltoniae



Cyclonema

Cephalopoda



x 0.5



orthocones



x 0.5

Manticeras



x 0.5

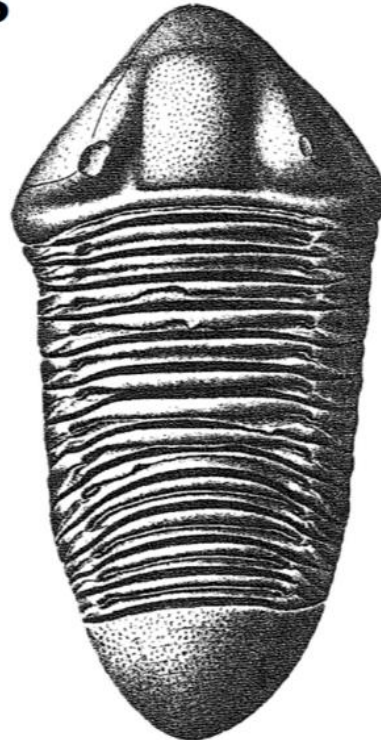
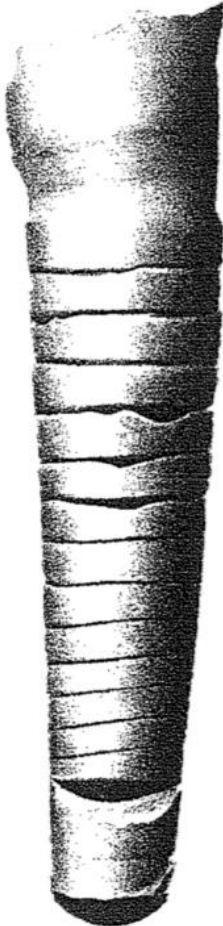
Agoniatites



trilobites and other arthropods

cephalopods

Michelinoceras telamon



Dipleura dekayi



Eldredgeops (Phacops) rana

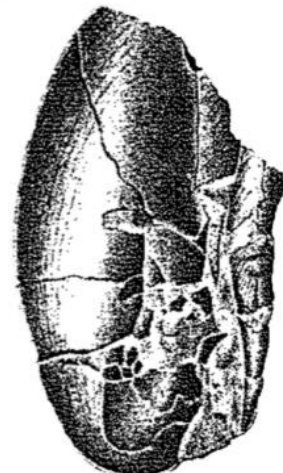


Greenops/Bellacarwrightia

Rhinocaris columbina: a rare type of crustacean known as a phyllocarid; only carapaces are found.

bryozoan

Leptotrypella amplexans: an encrusting bryozoan; at Geer Road it often is found on *Palaeozygopleura* gastropods





References

- Allmon, W.D. (2022a). Phylum Cnidaria [lecture]. In Allmon, W.D., *Paleobiology*. Cornell University, Ithaca, NY.
- Allmon, W.D. (2022b). Phylum Echinodermata [lecture]. In Allmon, W.D., *Paleobiology*. Cornell University, Ithaca, NY.
- Allmon, W.D. (2022c). The Cambrian Explosion [lecture]. In Allmon, W.D. & Fernandez, N.M., *Evolution of the Earth system*. Cornell University, Ithaca, NY.
- Allmon, W.D. (2022d). Trace fossils [lecture]. In Allmon, W.D., *Paleobiology*. Cornell University, Ithaca, NY.
- Allmon, W.D., & Mikkelsen, P.M. (2021). Class Gastropoda. In *The digital encyclopedia of ancient life*, <https://www.digitalatlasofancientlife.org/learn/mollusca/bivalvia/>
- Allmon, W.D., & Mikkelsen, P.M. (2020). Class Bivalvia. In *The digital encyclopedia of ancient life*, <https://www.digitalatlasofancientlife.org/learn/mollusca/bivalvia/>
- Benton, M.J., & Harper, D.A.T. (2020a). Ecdysozoa: arthropods. In Benton, M.J., & Harper, D.A.T., *Introduction to paleobiology and the fossil record* (2nd ed.) (pp. 413-441). Hoboken, NJ: Wiley-Blackwell.
- Benton, M.J., & Harper, D.A.T. (2020b). Lophophorates I: brachiopods and bryozoans. In Benton, M.J., & Harper, D.A.T., *Introduction to paleobiology and the fossil record* (2nd ed.) (pp. 345-369). Hoboken, NJ: Wiley-Blackwell.
- Benton, M.J., & Harper, D.A.T. (2020c). The basal metazoans: sponges and corals. In Benton, M.J., & Harper, D.A.T., *Introduction to paleobiology and the fossil record* (2nd ed.) (pp. 322-341). Hoboken, NJ: Wiley-Blackwell.
- Brett, C.E., & Baird, G.C. (2022). Middle Devonian environments of central and western New York State [field trip guidebook].
- Brett, C.E., Baird, G.C., Bartholomew, A.J., DeSantis, M.K., & Ver Straeten, C.A. (2012). Sequence stratigraphy and a revised sea-level curve for the Middle Devonian of eastern North America. *Palaeogeography, palaeoclimatology, palaeoclimate*, 304, 21-53. DOI: 10.1016/j.palaeo.2010.10.009
- Fernandez, N.M. (2022). The Devonian Foreland Basin of New York State and Pennsylvania: sedimentary basins, accumulation of minerals and organic matter, and transformation to oil and gas [lecture]. In Allmon, W.D. & Fernandez, N.M., *Evolution of the Earth system*. Cornell University, Ithaca, NY.
- Hendricks, J.R. (2018). Class Cephalopoda. In *The digital encyclopedia of ancient life*, <https://www.digitalatlasofancientlife.org/learn/mollusca/cephalopoda/>
- Hensen, C.J., & Carnevale, A.V. (2023). Section of the Windom Shale Coral Bed.
- Pier, J.Q. (n.d.). Symmetrical differences [figure]. In *The digital encyclopedia of ancient life*, <https://www.digitalatlasofancientlife.org/learn/brachiopoda/brachiopoda-vs-bivalvia/>
- Sang, S., Allmon, W.D., & Brett, C.E. (2013). Long before Cayuga's waters: the spectacular Devonian geology and paleontology of Ithaca, New York region. *Mid-America paleontology section digest*, 36(2), 74-92.
- Stanley, Jr., G.D. (2003). The evolution of modern corals and their early history. *Earth science reviews*, 60(3-4), 195-225. DOI: 10.1016/S0012-8252(02)00104-6
- Stokes, P.J., & Schreiber, H.A. (2017). Penn Dixie Fossil Park and Nature Preserve: a window



into the Devonian Period of western New York. *New York State Geological Association*, 89th Annual Meeting Guidebook, 1-18.

Welych-Flanagan, M. (n.d.). Common and interesting fossils found at Geer Road Quarry. *Paleontological Research Institute*.

Wilson, K.A. (2014). *Field guide to the Devonian fossils of New York*. Ithaca, NY: Paleontological Research Institution.